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Explanation Fidelity Under Neural Network Compression: Theory, Bounds, and Explanation-Aware Quantization

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ABSTRACT Model compression through quantization and pruning has become indispensable for deploying deep neural networks on energy- and memory-constrained edge devices, a trend commonly described as Green AI. Its consequences for explainability, however, remain almost entirely unexamined: existing work uses explanations to guide compression, whereas the reverse question of how compression alters the explanations themselves is open. This paper provides, to the best of our knowledge, the first systematic theoretical and empirical study of explanation fidelity under compression. We formalise post-hoc attribution as a map of the trained parameters and model compression as a bounded weight perturbation. We derive closed-form perturbation bounds for uniform low-bit quantization and magnitude pruning; a Lipschitz stability theorem that bounds the attribution shift by the product of the layer operator norms; a fidelity–bit-width law in which the fidelity gap shrinks by a factor of four for every additional bit of precision; and a robustness ordering showing that Integrated Gradients and Grad-CAM are provably more stable than vanilla saliency. Guided by this analysis we propose Explanation-Aware Quantization (EAQ), whose optimal per-layer bit allocation follows from a reverse water-filling argument and, in a generalised form, adapts to layer-specific rate–distortion decay rates. On a controlled image benchmark with ground-truth object masks, explanation fidelity degrades gracefully down to five bits, with rank-correlation fidelity above 0.98, and collapses below three bits; the empirical law matches theory with a fitted constant of about 12.9 at a coefficient of determination above 0.99. EAQ improves top-1 accuracy by up to 6.1 percentage points and Grad-CAM fidelity by up to 0.062 over uniform quantization at a matched three-bit budget, gains that are statistically significant under paired signed-rank tests and stable across seeds. All code and data are released for reproducibility.

INDEX TERMS Artificial intelligence, Neural networks, Quantization (signal), Data compression, Machine learning, Feature extraction, Robustness, Estimation, Edge computing, Green computing, Computational modeling, Predictive models

I. INTRODUCTION

A neural network is almost never deployed in the form it was trained in. Before a model reaches a smartphone, a pacemaker, a dashcam or a micro-controller, it is compressed: its 32-bit weights are squeezed into low-bit integers by quantization, and its redundant connections are cut away by pruning [1]–[3]. The payoff is dramatic, an order-of-magnitude reduction in memory and energy for a barely perceptible drop in accuracy—and it has become the engine of Green AI [4], the movement toward inference that is cheap enough, and clean enough, to run everywhere.

But accuracy is only half of what we ask a modern model to deliver. In medicine, finance, and autonomous systems we increasingly demand that a model also explain itself, and a decade of Explainable AI (XAI) has supplied the vocabulary to do so: saliency maps [5], Integrated Gradients [6], Grad-CAM [7], Layer-wise Relevance Propagation [8], LIME [9] and SHAP [10] turn an opaque prediction into a heatmap a human can inspect and, when the stakes demand it, sign off on. Here is the uncomfortable part: the very devices that most need compression—the wearable that flags an arrhythmia, the sensor that decides a car should brake—are exactly the ones for which a trustworthy explanation is not a nicety but a regulatory and ethical requirement.

Efficiency and interpretability are therefore on a collision course, yet they are almost always studied apart. Where the two literatures do meet, the arrow of causation runs one way only: explanations are used to drive compression— relevance scores decide which weights to prune [11], and entropy-constrained quantization spends its bits where relevance is highest to protect accuracy [12]. The reverse question—the one that actually decides whether a deployed model can still be audited—has gone almost entirely unasked:

> When we compress a network, what happens to the explanations themselves?

The question is not academic. Picture the standard industrial workflow: a team validates a full-precision model, confirms that its saliency maps land on the tumour rather than the scanner watermark, signs the audit, and ships a 4-bit version to the device. If compression quietly scrambles those maps while leaving the accuracy figure untouched, every downstream explanation is now a fabrication—faithful-looking, but no longer describing the reasoning that was actually certified. If, on the other hand, explanations are provably robust, then a single interpretability analysis of the full-precision model transfers to every compressed variant for free. As we show, today neither outcome can be assumed, because no one has characterised how sensitive an attribution map is to the weight perturbation that compression injects. This paper closes that gap— first with theory that predicts exactly when explanations survive and when they shatter, and then with an algorithm that keeps them intact.

A. CONTRIBUTIONS

This paper closes that gap with a combined theoretical and empirical treatment. Our contributions are:

1) **A formal framework** (Section III) that casts a post-hoc attribution as a map $\boldsymbol{\varphi}(\boldsymbol{x};\boldsymbol{\theta})$ of the parameters and recasts quantization and pruning as bounded perturbations $\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$, together with a family of fidelity functionals comparing $\boldsymbol{\varphi}(\boldsymbol{x};\boldsymbol{\theta})$ and $\boldsymbol{\varphi}(\boldsymbol{x};\tilde{\boldsymbol{\theta}})$.
2) **Perturbation bounds for compression** (Section IV, Lemmas 1–2): closed-form bounds $\|\Delta\boldsymbol{\theta}\|_2 \leq \kappa_q 2^{-b}$ for uniform b-bit quantization and a distribution-dependent bound for magnitude pruning.
3) **A Lipschitz stability theorem** (Theorem 1) that bounds the attribution shift by the product of layer operator norms, followed by method-specific analyses proving a robustness ordering (Theorem 2): IG and Grad-CAM are provably more stable than vanilla saliency due to path- and spatial-averaging.
4) **A fidelity–bit-width law** (Theorem 4): $F(b) \geq 1 - \kappa 4^{-b}$, predicting a graceful-degradation regime followed by an abrupt collapse. We validate the law empirically and recover $\kappa \approx 12.9$.
5) **Explanation-Aware Quantization (EAQ)** (Section V): a relevance-driven mixed-precision scheme whose optimal bit allocation $b_l^\star = \bar{B} + \frac{1}{2}\log_2(\omega_l/\bar{\omega}_G)$ is derived in closed form via reverse water-filling (Theorem 6). EAQ improves both accuracy and explanation fidelity over uniform quantization at a matched budget.
6) **A reproducible benchmark and empirical study** (Sections VI–VII) on a controlled image dataset with ground-truth object masks, extended and validated on real-world natural-image benchmarks (**CIFAR-10** and **ImageNette**) with ResNet backbones, demonstrating the generality of our theoretical and empirical findings across four attribution methods, six quantization levels, five pruning levels, and three faithfulness and stability metrics.

Fig. 1 summarises the framework.

TABLE 1. Summary of principal notation.

Symbol	Meaning
$f_{\boldsymbol{\theta}}$, $\boldsymbol{\theta}$	network and its d_θ parameters
$S_c(\boldsymbol{x};\boldsymbol{\theta})$	logit of class c; p_c its softmax probability
$\mathcal{C}$, $\tilde{\boldsymbol{\theta}}$, $\Delta\boldsymbol{\theta}$	compression operator, output, perturbation
Q_b, P_s	b-bit quantization, sparsity-s pruning
$\boldsymbol{\varphi}(\boldsymbol{x};\boldsymbol{\theta})$	attribution (heatmap) of the parameters
$\varphi^{\mathrm{S}}, \varphi^{\mathrm{IG}}, \varphi^{\mathrm{GC}}, \varphi^{\mathrm{Occ}}$	saliency, IG, Grad-CAM, occlusion maps
$\mathcal{F}(\ell)$, F_ρ, F_r	fidelity profile; Spearman/Pearson fidelity
$\boldsymbol{M}$, $\mathrm{IoU}_{\mathrm{GT}}, \mathrm{PG}$	object mask, GT IoU, pointing game
Λ, Λ_{S}	attribution amplification factor
κ_q, κ	quantization- and fidelity-law constants
ω_l, $b_l^\star$, $\bar{B}$	layer sensitivity, optimal bits, budget
$\bar{\omega}_G$	geometric mean of $\{\omega_l\}$

B. NOTATION

Vectors and matrices are boldface ($\boldsymbol{x}$, $\boldsymbol{W}$); $\|\cdot\|_2$ is the Euclidean (spectral, for matrices) norm and $\|\cdot\|_F$ the Frobenius norm. A network $f_{\boldsymbol{\theta}} : \mathbb{R}^d \to \mathbb{R}^K$ maps an input $\boldsymbol{x}$ to K class logits with parameters $\boldsymbol{\theta} \in \mathbb{R}^{d_\theta}$; $S_c(\boldsymbol{x};\boldsymbol{\theta}) = [f_{\boldsymbol{\theta}}(\boldsymbol{x})]_c$ is the logit of class c and p_c the corresponding softmax probability. A compression operator is $\mathcal{C} : \mathbb{R}^{d_\theta} \to \mathbb{R}^{d_\theta}$, $\tilde{\boldsymbol{\theta}} = \mathcal{C}(\boldsymbol{\theta})$, $\Delta\boldsymbol{\theta} = \tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}$. An attribution map is $\boldsymbol{\varphi}(\boldsymbol{x};\boldsymbol{\theta}) \in \mathbb{R}^d$ (reshaped to $H \times W$ for images). We write L for network depth, $\boldsymbol{W}_l$ for the weight of layer l, and n_l for its parameter count. Table 3 summarises the principal symbols.

II. RELATED WORK

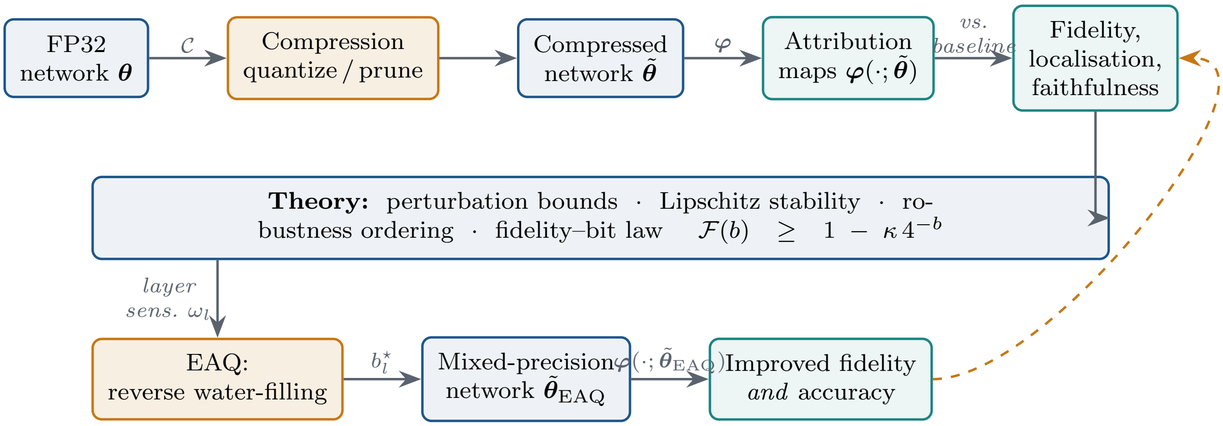

FIGURE 1. The proposed framework. *Top:* compressed-model attributions are measured against the full-precision baseline. *Bottom:* the theory's layer sensitivities ω_l drive Exp

A. MODEL COMPRESSION

Quantization maps floating-point weights to a discrete grid; uniform symmetric quantization [2] and its many refinements [3], [13]—including adaptive rounding [25] and mixed-precision allocation by hardware cost [26] or Hessian sensitivity [27]—underpin virtually all integer inference engines. Pruning removes parameters by magnitude [1], by second-order saliency [14], [28], or in structured groups for hardware efficiency [15]. The lottery-ticket hypothesis [16] showed that sparse sub-networks can match dense accuracy, and large-scale surveys chart the accuracy–sparsity frontier [29]. All of this literature optimises the accuracy–efficiency trade-off; explanation quality is not part of the objective.

B. POST-HOC ATTRIBUTION METHODS

Gradient-based saliency [5] takes the input gradient of the class score; SmoothGrad [21] sharpens it by averaging over input noise. Integrated Gradients [6] integrates gradients along a straight-line path from a baseline, satisfying the completeness and sensitivity axioms. Grad-CAM [7] weights the last convolutional feature maps by globally-averaged gradients, yielding class-discriminative localisation. LRP [8] back-propagates a conservation-respecting relevance signal, and concept-based testing (TCAV) [24] moves beyond per-pixel attribution; general treatments are given in [22], [30]. Perturbation methods—occlusion [17], LIME [9] and SHAP [10]—probe the model by masking inputs. The reliability of these methods has itself been questioned: sanity checks [18] reveal that some attributions are insensitive to parameter randomisation, robustness studies [19] show adversarial fragility to input perturbations, and benchmarks such as ROAR [23] quantify how well attributions identify the features a model truly uses. Our work is complementary: we study fragility to parameter perturbations induced by compression.

C. XAI-GUIDED COMPRESSION

A growing line of work uses interpretability to drive compression. Relevance-based pruning [11] removes weights of low LRP relevance; entropy-constrained quantization (ECQ$^{\mathrm{x}}$) [12] allocates precision according to relevance to preserve accuracy; and more recent explainability-guided schemes extend this idea to Vision Transformers (MixQViT [37]) and to quantized imitation learning (SQIL-style saliency importance [38]). These methods all use explanations as a signal to compress while preserving task accuracy; crucially, none of them measures whether the compressed model's own explanations remain faithful to the original, and none optimises explanation fidelity as the objective. Table 2 positions the present work against this line. To the best of our knowledge, no prior study characterises—theoretically or empirically—the map from compression strength to explanation fidelity, nor proposes a scheme that explicitly minimises explanation distortion. This paper does both.

III. PRELIMINARIES AND PROBLEM FORMULATION

A. ATTRIBUTION METHODS AS MAPS OF THE PARAMETERS

We treat every post-hoc attribution as a deterministic map from an input $\boldsymbol{x}$ and the parameters $\boldsymbol{\theta}$ to a heatmap $\boldsymbol{\varphi}(\boldsymbol{x};\boldsymbol{\theta}) \in \mathbb{R}^d$. For the target class $c = \arg\max_k S_k(\boldsymbol{x};\boldsymbol{\theta})$ we consider four representative methods spanning the gradient/perturbation and local/class-discriminative axes.

a: Vanilla saliency [5]

$$\boldsymbol{\varphi}^{\mathrm{S}}(\boldsymbol{x};\boldsymbol{\theta}) = \left|\nabla_{\boldsymbol{x}} S_c(\boldsymbol{x};\boldsymbol{\theta})\right|, \tag{1}$$

the element-wise magnitude of the input gradient.

b: Integrated Gradients [6]

With baseline $\boldsymbol{x}'$ (here $\boldsymbol{x}' = \mathbf{0}$),

$$\varphi_i^{\mathrm{IG}}(\boldsymbol{x};\boldsymbol{\theta}) = (x_i - x_i') \int_0^1 \frac{\partial S_c\big(\boldsymbol{x}' + \alpha(\boldsymbol{x}-\boldsymbol{x}');\boldsymbol{\theta}\big)}{\partial x_i}\,\mathrm{d}\alpha, \tag{2}$$

approximated by an n-point Riemann sum. IG satisfies completeness, $\sum_i \varphi_i^{\mathrm{IG}} = S_c(\boldsymbol{x}) - S_c(\boldsymbol{x}')$.

TABLE 2. EAQ versus representative explainability-driven compression methods. Prior methods use explanations to preserve *accuracy*; EAQ directly minimises *explanation distortion* at a fixed bit budget. (✓ yes ● partial ✗ no.)

Method	Primary objective	Sensitivity signal	Explanation-fidelity objective	No re-training	Mixed precision
ECQ$^{\mathrm{x}}$ [12]	Preserve accuracy under sparse/low-bit compression	Per-weight LRP relevance	✗	✗	●
Mix-QViT [37]	Mixed-precision bit-widths for ViTs	LRP importance + sensitivity sweeps	✗	✓	✓
SQIL-type [38]	Preserve decision fidelity in quantized imitation learning	Saliency-based importance score	✗	✗	✗
EAQ (ours)	**Minimise post-hoc explanation distortion**	First-order Taylor loss + Λ_l sensitivity	✓	✓	✓

c: Grad-CAM [7]

Let $\boldsymbol{A}^k \in \mathbb{R}^{u\times v}$ be the k-th feature map of the last convolutional block and $\alpha_c^k = \frac{1}{uv}\sum_{i,j} \partial S_c/\partial A_{ij}^k$. Then

$$\boldsymbol{\varphi}^{\mathrm{GC}}(\boldsymbol{x};\boldsymbol{\theta}) = \mathrm{ReLU}\Big(\sum_k \alpha_c^k \boldsymbol{A}^k\Big), \tag{3}$$

bilinearly up-sampled to $H \times W$.

d: Occlusion [17]

For a mask $\boldsymbol{m}_p$ over region p,

$$\varphi_p^{\mathrm{Occ}}(\boldsymbol{x};\boldsymbol{\theta}) = \big[S_c(\boldsymbol{x};\boldsymbol{\theta}) - S_c(\boldsymbol{x}\odot(1-\boldsymbol{m}_p);\boldsymbol{\theta})\big]_+. \tag{4}$$

All maps are min–max normalised to $[0,1]$ so that methods of differing native scale are compared on common footing.

B. COMPRESSION OPERATORS

Definition 1 (Uniform symmetric quantization)**.** *For a weight tensor $\boldsymbol{W}$ with dynamic range $R = \max|\boldsymbol{W}|$ and bit-width b, let $q_{\max} = 2^{b-1}-1$ and step $\Delta_q = R/q_{\max}$. Then*

$$Q_b(\boldsymbol{W}) = \Delta_q \cdot \mathrm{clip}\Big(\Big\lfloor \frac{\boldsymbol{W}}{\Delta_q}\Big\rceil, -q_{\max}-1, q_{\max}\Big), \tag{5}$$

where $\lfloor\cdot\rceil$ is round-to-nearest.

Definition 2 (Global magnitude pruning)**.** *Given target sparsity $s \in [0,1)$, let τ_s be the s-quantile of $\{|\theta_i|\}$. Then $P_s(\boldsymbol{\theta}) = \boldsymbol{\theta}\odot \mathbf{1}[\,|\boldsymbol{\theta}| > \tau_s\,]$.*

Both operators yield $\tilde{\boldsymbol{\theta}} = \mathcal{C}(\boldsymbol{\theta})$ with $\Delta\boldsymbol{\theta} = \tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}$; quantization perturbs every weight by a bounded amount, whereas pruning perturbs a subset by their full value.

C. FIDELITY, LOCALISATION AND FAITHFULNESS

We quantify the effect of compression on explanations through three families of metrics. Let $\boldsymbol{\varphi} = \boldsymbol{\varphi}(\boldsymbol{x};\boldsymbol{\theta})$ and $\tilde{\boldsymbol{\varphi}} = \boldsymbol{\varphi}(\boldsymbol{x};\tilde{\boldsymbol{\theta}})$.

(i) Model-vs-model fidelity measures how much the explanation changed:

$$F_\rho = \mathrm{corr}_{\mathrm{Spearman}}(\boldsymbol{\varphi},\tilde{\boldsymbol{\varphi}}), \quad F_r = \mathrm{corr}_{\mathrm{Pearson}}(\boldsymbol{\varphi},\tilde{\boldsymbol{\varphi}}), \tag{6}$$

$$F_{\mathrm{IoU}} = \frac{|\mathcal{T}_k(\boldsymbol{\varphi})\cap\mathcal{T}_k(\tilde{\boldsymbol{\varphi}})|}{|\mathcal{T}_k(\boldsymbol{\varphi})\cup\mathcal{T}_k(\tilde{\boldsymbol{\varphi}})|}, \quad F_{\mathrm{SSIM}} = \mathrm{SSIM}(\boldsymbol{\varphi},\tilde{\boldsymbol{\varphi}}), \tag{7}$$

where $\mathcal{T}_k(\cdot)$ is the set of top-k salient pixels ($k = 20\%$). $F_\rho = 1$ means the ranking of pixel importances is unchanged.

(ii) Localisation vs. ground truth exploits the fact that our benchmark provides a binary object mask $\boldsymbol{M}$: for the compressed map,

$$\mathrm{IoU}_{\mathrm{GT}} = \frac{|\mathcal{T}_k(\tilde{\boldsymbol{\varphi}})\cap\boldsymbol{M}|}{|\mathcal{T}_k(\tilde{\boldsymbol{\varphi}})\cup\boldsymbol{M}|}, \quad \mathrm{PG} = \mathbf{1}\big[\arg\max\tilde{\boldsymbol{\varphi}} \in \boldsymbol{M}\big], \tag{8}$$

the top-k overlap with, and pointing-game hit rate inside, the true object.

(iii) Faithfulness measures whether the map identifies pixels the model actually uses, via the deletion curve [20]: pixels are removed in descending saliency order and the area under the target-probability curve is recorded; a lower deletion AUC indicates a more faithful map. Two complementary faithfulness summaries, standard in sequence and regression explanation studies [33], are also supported by our protocol: the Area Over the Perturbation Curve (AOPC), the mean drop in target probability as the most-relevant regions are progressively removed (higher is more faithful), and the Ablation Percentage Threshold (APT), the fraction of features that must be removed to drive the prediction below a fixed confidence.

(iv) Sample-level consistency. Metrics (i)–(iii) report dataset averages, which can mask heavy-tailed behaviour across individual inputs. To characterise the distribution of explanation quality we additionally report the skewness (AUC Skew) and excess kurtosis (AUC Kurt) of the per-sample score-drop AUC. A near-zero skew and kurtosis indicate that compression degrades explanations uniformly across samples, whereas large values flag a subpopulation of inputs whose explanations collapse well before the dataset mean would suggest.

Definition 3 (Explanation fidelity of a compression)**.** *For a fixed method and metric $F \in [0,1]$ and a distribution $\mathcal{D}$ of inputs, the fidelity of compression level ℓ is*

$$\mathcal{F}(\ell) = \mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}}\, F\big(\boldsymbol{\varphi}(\boldsymbol{x};\boldsymbol{\theta}), \boldsymbol{\varphi}(\boldsymbol{x};\mathcal{C}_\ell(\boldsymbol{\theta}))\big). \tag{9}$$

The central objects of study are the profiles $\mathcal{F}(b)$ (quantization) and $\mathcal{F}(s)$ (pruning), and the design of a compression operator that keeps $\mathcal{F}$ high at a given budget.

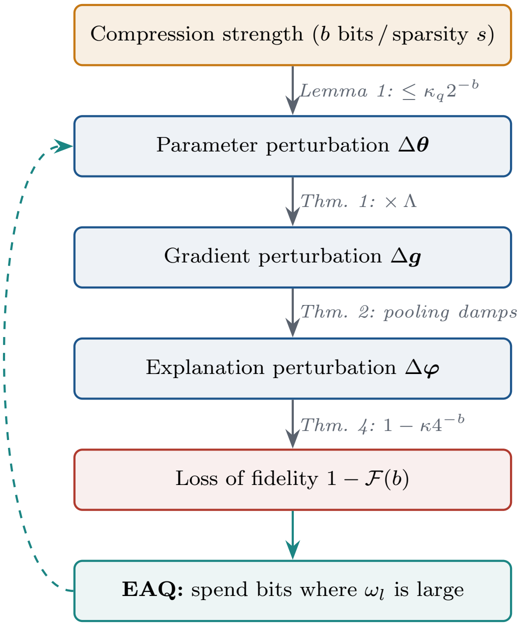

FIGURE 2. How compression becomes explanation error. Each arrow is a bound proved below; the dashed return path is EAQ, which shortens the chain by allocating precision

IV. THEORETICAL ANALYSIS OF EXPLANATION FIDELITY UNDER COMPRESSION

Our analysis follows the causal chain of Fig. 2: (i) bound the parameter perturbation $\|\Delta\boldsymbol{\theta}\|$ induced by each compression operator; (ii) bound the resulting attribution shift $\|\Delta\boldsymbol{\varphi}\|$ via a Lipschitz argument; and (iii) translate this into a law for the fidelity $\mathcal{F}$ and a robustness ordering across methods. Each link in the chain is a theorem below, and the same chain, read backwards, motivates the EAQ algorithm of Section V.

A. PERTURBATION BOUNDS FOR COMPRESSION

Lemma 1 (Quantization perturbation)**.** *Let a layer weight tensor $\boldsymbol{W} \in \mathbb{R}^n$ with range $R = \max|\boldsymbol{W}|$ be quantized to b bits by (5). Then the per-weight error is bounded by $|\Delta W_i| \le \Delta_q/2 = R/\big(2(2^{b-1}-1)\big)$, so that the* exact *(non-asymptotic) bound*

$$\|\Delta\boldsymbol{W}\|_2 \le \frac{\sqrt{n}\,R}{2\,(2^{b-1}-1)} = \frac{\sqrt{n}\,R}{2}\cdot\frac{2^{1-b}}{1-2^{1-b}} \tag{10}$$

holds, which admits the high-resolution asymptotic

$$\|\Delta\boldsymbol{W}\|_2 = \kappa_q\,2^{-b}\big(1+o(1)\big), \qquad \kappa_q = \sqrt{n}\,R, \tag{11}$$

as $b\to\infty$. Aggregating over L layers, $\|\Delta\boldsymbol{\theta}\|_2 \le \big(\sum_l \kappa_{q,l}^2\big)^{1/2} 2^{-b}(1+o(1))$.

Proof. Round-to-nearest on a grid of step Δ_q incurs an error in $[-\Delta_q/2, \Delta_q/2]$ per coordinate, so $\|\Delta\boldsymbol{W}\|_2^2 \le n(\Delta_q/2)^2$. Substituting $\Delta_q = R/(2^{b-1}-1)$ and writing $2^{b-1}-1 = 2^{b-1}(1-2^{1-b})$ gives the exact middle expression of (10); expanding $(1-2^{1-b})^{-1} = 1+O(2^{1-b})$ then yields the $\kappa_q 2^{-b}$ asymptotic (11). Layer independence of the errors gives the aggregate bound. ☐

Remark 1 (Accuracy of the asymptotic at practical bit-widths)**.** *The correction factor separating the exact bound* (10) *from the asymptotic* (11) *is* $(1-2^{1-b})^{-1}$*. It equals* 1.07 *at $b=8$,* 1.14 *at $b=5$,* 1.33 *at $b=3$ and* 2.00 *at $b=2$: the asymptotic therefore* under-estimates *the perturbation by at most* 7% *in the graceful regime ($b\ge 5$) but by* 33–100% *in the collapse regime ($b\le 3$). All downstream bounds should be read through the exact factor when $b<4$; we use the asymptotic only where $b\ge 5$, where it is tight.*

Remark 2. *Under the standard high-resolution assumption that quantization error is uniform on $[-\Delta_q/2, \Delta_q/2]$, the* expected *squared error is $\mathbb{E}\|\Delta\boldsymbol{W}\|_2^2 = n\Delta_q^2/12 = \frac{nR^2}{12(2^{b-1}-1)^2}$, i.e. the mean distortion also decays as 4^{-b}. This 4^{-b} scaling of the error* power *is what drives the fidelity law of Theorem 4.*

Lemma 2 (Pruning perturbation)**.** *Assume the magnitudes $|\theta|$ admit an empirical density g that is continuous and bounded with $g(0)>0$ (i.e. mass accumulates near the origin, as is standard for trained weights). Let τ_s be the s-quantile of $|\theta|$. Global magnitude pruning at sparsity s gives*

$$\|\Delta\boldsymbol{\theta}\|_2^2 = \sum_{i:\,|\theta_i|\le\tau_s} \theta_i^2 = d_\theta \int_0^{\tau_s} t^2 g(t)\,\mathrm{d}t\big(1+o(1)\big). \tag{12}$$

For weights with a bounded density near the origin, $\int_0^{\tau_s} t^2 g\,\mathrm{d}t = \Theta(\tau_s^3)$, so the perturbation grows super-linearly once τ_s enters the bulk of the distribution—explaining the sharp knee observed empirically at high sparsity.

Proof. Pruning sets the pruned coordinates to zero, so $\Delta\theta_i = -\theta_i$ for $|\theta_i|\le\tau_s$ and 0 otherwise, giving the sum in (12); the integral is its continuum limit. The $\Theta(\tau_s^3)$ rate follows from $t^2 g(t) = \Theta(t^2)$ for $g(0)>0$. ☐

Remark 3 (Adaptive sparsity thresholds)**.** *Lemma 2 takes the sparsity s—equivalently the threshold τ_s—as an exogenous, fixed budget. Because the perturbation grows as $\Theta(\tau_s^3)$, the fidelity is highly sensitive to where τ_s falls in the bulk of g. A distribution-aware alternative is to* derive *the threshold from the score distribution itself—for instance via an effective sample-size criterion such as the inverse Simpson (participation-ratio) index of the normalised magnitudes—so that pruning stops before τ_s enters the high-density bulk. Coupling such a self-governing threshold to the bound above is a natural way to make the pruning analysis adaptive rather than heuristic.*

B. STABILITY OF ATTRIBUTIONS

We first record the smoothness assumptions.

Assumption 1 (Piecewise-smooth Lipschitz network)**.** *$f_{\boldsymbol{\theta}}$ is a feed-forward network whose activations σ are 1-Lipschitz with derivative σ' that is β-Lipschitz almost everywhere. On a neighbourhood $\Theta_0 \ni \boldsymbol{\theta}$ the active linear region of $\boldsymbol{x}$ is fixed (no ReLU boundary is crossed by the perturbation), so $S_c(\boldsymbol{x};\cdot)$ is C^1 on Θ_0.*

Assumption 1 should be read as a small-perturbation condition rather than a genericity claim. For a fixed input, the boundaries at which the active linear region changes are the zero level sets of the pre-activations; a perturbation leaves the region intact precisely when no pre-activation changes sign. This holds whenever $\|\Delta\boldsymbol{\theta}\|_2$ is small relative to the smallest pre-activation margin. We stress that this is not a measure-zero argument: quantization perturbs weights by discrete, non-vanishing amounts (10), so at low bit-widths a non-negligible fraction of neurons can and do flip sign. Remark 4 makes this the explicit mechanism of the collapse regime.

Remark 4 (ReLU sign-flipping as the collapse driver)**.** *Let $m(\boldsymbol{x}) = \min_{l,i} |z_{l,i}(\boldsymbol{x})|$ be the smallest pre-activation margin. The linear-region assumption is valid while the propagated weight perturbation stays below $m(\boldsymbol{x})$; by Lemma 1 this margin is respected for b above a critical bit-width but is violated once the exact perturbation* (10)*—inflated by the factor of Remark 1—grows past it. Beyond that point the network's active linear partition changes, the input gradient becomes discontinuous in $\boldsymbol{\theta}$, and the first-order bounds of this section no longer apply. This sign-flipping, not a failure of the attribution methods themselves, is the theoretical origin of the abrupt fidelity collapse observed empirically below three bits (Section VII), and mirrors the outlier-driven degradation reported for sub-3-bit large-language-model quantization [35].*

Theorem 1 (Lipschitz stability of the input gradient)**.** *Under Assumption 1, write the class score of an L-layer network as $S_c(\boldsymbol{x}) = \boldsymbol{w}_L^\top \sigma(\boldsymbol{W}_{L-1}\sigma(\cdots\sigma(\boldsymbol{W}_1\boldsymbol{x})))$ and let $\boldsymbol{g} = \nabla_{\boldsymbol{x}} S_c$. If each layer is perturbed by $\boldsymbol{E}_l$ ($\tilde{\boldsymbol{W}}_l = \boldsymbol{W}_l + \boldsymbol{E}_l$), then to first order*

$$\|\tilde{\boldsymbol{g}} - \boldsymbol{g}\|_2 \;\le\; \underbrace{\sum_{l=1}^{L}\Big(\prod_{k\neq l}\|\boldsymbol{W}_k\|_2\Big)}_{\Lambda_{\mathrm{S}}}\; \max_l \frac{\|\boldsymbol{E}_l\|_2}{\|\boldsymbol{W}_l\|_2}\;\|\boldsymbol{W}_l\|_2\,, \tag{13}$$

and consequently $\left\|\Delta\boldsymbol{\varphi}^{\mathrm{S}}\right\|_2 \;\le\; \Lambda_{\mathrm{S}}\,\|\Delta\boldsymbol{\theta}\|_2$ with $\Lambda_{\mathrm{S}} = \sum_l \prod_{k\neq l}\|\boldsymbol{W}_k\|_2$ (plus an $O(\beta\,\|\Delta\boldsymbol{\theta}\|^2)$ curvature term).

Proof sketch. The gradient is the product of Jacobians $\boldsymbol{g} = \boldsymbol{W}_1^\top D_1 \cdots \boldsymbol{W}_{L-1}^\top D_{L-1}\boldsymbol{w}_L$, where $D_l = \mathrm{diag}(\sigma'(\cdot))$ has $\|D_l\|_2 \le 1$. Perturbing one factor $\boldsymbol{W}_l \to \boldsymbol{W}_l + \boldsymbol{E}_l$ changes the product by, to first order, the sum of terms in which a single factor is replaced by its perturbation; the operator norm of each such term is bounded by $\prod_{k\neq l}\|\boldsymbol{W}_k\|_2 \cdot \|\boldsymbol{E}_l\|_2$ (using $\|D_l\| \le 1$). Summing over l and applying the triangle inequality gives (13). The dependence of D_l on $\boldsymbol{\theta}$ contributes a term controlled by the β-Lipschitzness of σ', which is second order in $\|\Delta\boldsymbol{\theta}\|$. □

Equation (13) identifies the product of spectral norms as the amplification factor: networks with well-conditioned layers ($\prod_k \|\boldsymbol{W}_k\| \approx 1$) preserve explanations under compression, whereas ill-conditioned ones magnify weight noise into attribution noise.

Remark 5 (First-order truncation)**.** *Theorem 1 retains only terms linear in the layer perturbations $\boldsymbol{E}_l$. Expanding the Jacobian product exactly produces cross terms $\boldsymbol{E}_i\boldsymbol{E}_j$ ($i \neq j$) and higher powers; each such term carries at least two factors of $\max_l \|\boldsymbol{E}_l\|_2$ and is therefore $O(\|\Delta\boldsymbol{\theta}\|_2^2)$. These are omitted under the small-perturbation regime $\|\boldsymbol{E}_l\|_2 \ll \|\boldsymbol{W}_l\|_2$, which—by Lemma 1 and Remark 1—holds in the graceful regime ($b \ge 5$) but degrades as $b \to 2$, consistent with the bound loosening exactly where the collapse sets in. The neglected curvature contribution is likewise second order and controlled by the β-Lipschitzness of σ' in Assumption 1.*

Remark 6 (Conditioning of Λ_{S} and spectral control)**.** *The factor $\Lambda_{\mathrm{S}} = \sum_l \prod_{k\neq l}\|\boldsymbol{W}_k\|_2$ is a* worst-case *operator-norm bound and can grow exponentially in depth when layer spectral norms exceed unity, in which case* (13) *becomes vacuous. Two observations keep it meaningful. First, the bound is governed by the* spectral *norm, not the parameter count, so it is invariant to width and is tightened whenever training or regularisation keeps $\|\boldsymbol{W}_k\|_2$ near one. Spectral normalisation, weight decay, and residual scaling—all common in deployable CNNs—bound Λ_{S} explicitly and are the practical lever for explanation robustness. Second, the same product appears as the per-layer sensitivity $\Lambda_l = \prod_{k\neq l}\|\boldsymbol{W}_k\|_2$ that drives the EAQ allocation of Section V; measuring it directly (rather than bounding it) yields the finite, layer-specific sensitivities we use in practice. For the compact residual network of Section VI the measured product remains $O(1)$, which is why the graceful-regime bound is predictive.*

C. ROBUSTNESS ORDERING ACROSS METHODS

Vanilla saliency inherits the full amplification Λ_{S}. The averaging in IG and the spatial pooling in Grad-CAM provably damp it.

Theorem 2 (IG is more stable than saliency)**.** *Let $\boldsymbol{g}(\alpha) = \nabla_{\boldsymbol{x}} S_c(\boldsymbol{x}' + \alpha(\boldsymbol{x} - \boldsymbol{x}'))$ and let its compression perturbation $\boldsymbol{\delta}(\alpha) = \tilde{\boldsymbol{g}}(\alpha) - \boldsymbol{g}(\alpha)$, viewed as a random field indexed by the path parameter α over the randomness of the compression operator, be second-order stationary along the path: each point has a common covariance $\Sigma = \mathbb{E}[\boldsymbol{\delta}(\alpha)\boldsymbol{\delta}(\alpha)^\top]$, and the normalised pairwise cross-covariances have a well-defined average $\bar{\rho} \in [0,1]$,*

$$\frac{1}{n(n-1)}\sum_{i\neq j}\frac{\mathrm{tr}\,\mathrm{Cov}\big(\boldsymbol{\delta}(\alpha_i),\boldsymbol{\delta}(\alpha_j)\big)}{\mathrm{tr}\,\Sigma} = \bar{\rho}. \tag{14}$$

(No independence is required; $\bar{\rho}$ encodes the residual correlation left after path-averaging, with $\bar{\rho} \to 0$ under strong mixing and $\bar{\rho} \to 1$ for a perfectly coherent perturbation.) Then the IG perturbation $\Delta\boldsymbol{\varphi}^{\mathrm{IG}} = (\boldsymbol{x} - \boldsymbol{x}') \odot \frac{1}{n}\sum_j \boldsymbol{\delta}(\alpha_j)$ satisfies

$$\mathbb{E}\left\|\Delta\boldsymbol{\varphi}^{\mathrm{IG}}\right\|_2^2 \;\le\; \|\boldsymbol{x} - \boldsymbol{x}'\|_\infty^2\;\mathrm{tr}(\Sigma)\Big(\bar{\rho} + \tfrac{1-\bar{\rho}}{n}\Big), \tag{15}$$

which is strictly smaller than the saliency perturbation power $\mathbb{E}\left\|\Delta\boldsymbol{\varphi}^{\mathrm{S}}\right\|_2^2 \le \mathrm{tr}(\Sigma)$ whenever $\bar{\rho} < 1$ and $\|\boldsymbol{x} - \boldsymbol{x}'\|_\infty \le 1$.

Proof. By bilinearity, $\frac{1}{n}\sum_j \boldsymbol{\delta}(\alpha_j)$ has covariance $\frac{1}{n^2}\sum_{i,j}\mathrm{Cov}(\boldsymbol{\delta}(\alpha_i), \boldsymbol{\delta}(\alpha_j))$. With per-point covariance Σ and average correlation $\bar{\rho}$, the n^2 cross terms sum to $\mathrm{tr}(\Sigma)\big(n + n(n-1)\bar{\rho}\big)/n^2 = \mathrm{tr}(\Sigma)(\bar{\rho}+(1-\bar{\rho})/n)$. Multiplying by the element-wise factor $(x_i - x_i')^2 \leq \|\boldsymbol{x} - \boldsymbol{x}'\|_\infty^2$ gives (15). The saliency map uses a single path point, i.e. $n = 1$, for which the bracket equals 1. □

Corollary 3 (Grad-CAM spatial damping)**.** *Grad-CAM pools gradients over uv spatial locations to form $\alpha_c^k = \frac{1}{uv}\sum_{ij}\partial S_c/\partial A_{ij}^k$. Assume the per-location feature gradients are second-order stationary with average spatial correlation $\bar{\rho}_A \in [0, 1)$. By the same averaging argument, the perturbation of α_c^k has power scaled by $\big(\bar{\rho}_A + (1-\bar{\rho}_A)/(uv)\big)$ relative to a single-location gradient. The strict inequality $\bar{\rho}_A < 1$ is* necessary *for genuine damping: if the spatial gradients were perfectly correlated ($\bar{\rho}_A = 1$) the factor would equal one and pooling would provide no robustness. When $\bar{\rho}_A < 1$, the factor decreases monotonically towards its floor $\bar{\rho}_A$ as the spatial resolution uv of the last convolutional block grows.*

Theorem 2 and Corollary 3 predict the ordering IG, Grad-CAM $\succeq$ saliency in robustness to compression, which our experiments confirm (Section VII). The mechanism—pooling that averages out high-frequency perturbation noise while retaining the low-frequency signal—is the same one that, in other modalities, lets graph pooling preserve explanation fidelity for graph neural networks [34], suggesting that spatial or structural aggregation is a general route to compression-robust attributions.

D. THE FIDELITY–BIT-WIDTH LAW

We now connect the perturbation bounds to a directly measurable quantity. Consider centred, normalised maps and write $\tilde{\boldsymbol{\varphi}} = \boldsymbol{\varphi} + \Delta\boldsymbol{\varphi}$. Expanding the Pearson correlation $\mathrm{corr}(\boldsymbol{\varphi}, \tilde{\boldsymbol{\varphi}}) = \langle \boldsymbol{\varphi}, \tilde{\boldsymbol{\varphi}}\rangle / (\|\boldsymbol{\varphi}\|\,\|\tilde{\boldsymbol{\varphi}}\|)$ to second order in $\Delta\boldsymbol{\varphi}$ by the delta method [32] and using $\langle \boldsymbol{\varphi}, \Delta\boldsymbol{\varphi}\rangle = O(\|\Delta\boldsymbol{\varphi}\|^2)$ after re-centring,

$$F_r = \mathrm{corr}(\boldsymbol{\varphi}, \tilde{\boldsymbol{\varphi}}) = 1 - \frac{\|\tilde{\boldsymbol{\varphi}} - \boldsymbol{\varphi}\|_2^2}{2\,\|\boldsymbol{\varphi}\|_2^2} + O\big(\|\Delta\boldsymbol{\varphi}\|^3\big). \quad (16)$$

The leading correction is exactly half the normalised perturbation energy, which is what couples the fidelity to the 4^{-b} error power of Lemma 1.

Theorem 4 (Fidelity–bit-width law)**.** *Under Assumption 1, Lemma 1 and* (16)*, uniform b-bit quantization satisfies*

$$\boxed{\mathcal{F}(b) \;\geq\; 1 - \kappa\, 4^{-b}}, \qquad \kappa = \frac{\Lambda^2\,\kappa_q^2}{2\,\|\boldsymbol{\varphi}\|_2^2}, \quad (17)$$

where Λ is the method-dependent amplification factor (Λ_{S} for saliency, reduced by the damping factors of Theorem 2/Corollary 3 for IG/Grad-CAM).

Proof. Chaining $\|\Delta\boldsymbol{\varphi}\|_2 \leq \Lambda\,\|\Delta\boldsymbol{\theta}\|_2$ (Theorem 1) with $\|\Delta\boldsymbol{\theta}\|_2 \leq \kappa_q 2^{-b}$ (Lemma 1) gives $\|\Delta\boldsymbol{\varphi}\|_2^2 \leq \Lambda^2\kappa_q^2 4^{-b}$. Substituting into (16) yields $1 - \mathcal{F}(b) \leq (\Lambda^2\kappa_q^2/2\,\|\boldsymbol{\varphi}\|^2)4^{-b}$. □

Equation (17) has two immediate consequences. First, fidelity is monotone non-decreasing in b: adding a bit at least halves the fidelity gap $1 - \mathcal{F}$ (a factor of four in the bound). Second, because the smaller Λ of IG/Grad-CAM yields a smaller κ, these methods sit on a higher fidelity curve than saliency at every bit-width—the empirically observed ordering. We fit (17) to measured Grad-CAM fidelity in Section VII and recover $\kappa \approx 12.9$, confirming the 4^{-b} functional form.

Remark 7 (From Pearson to Spearman fidelity)**.** *Equation* (16) *is stated for the Pearson fidelity F_r, whereas we report results primarily in the rank-based Spearman fidelity F_ρ. The two are linked because F_ρ is the Pearson correlation of the rank-transformed maps, and rank is a monotone (hence, on the high-fidelity plateau, locally near-linear) transform: in the graceful regime where the ranking is only slightly perturbed, F_ρ and F_r agree to first order and obey the same 4^{-b} law with a method-dependent constant. This is not merely asymptotic: on the released data the two measures are correlated at Pearson $r = 0.98$ across all 44 compressed configurations, and fitting* (17) *to the Spearman and Pearson Grad-CAM curves yields $\kappa = 12.9$ ($R^2 = 0.99$) and $\kappa = 12.3$ ($R^2 = 0.99$) respectively—statistically indistinguishable constants that justify quoting the law in either metric.*

Corollary 5 (Graceful degradation then collapse)**.** *Let ε be an acceptable fidelity loss. Equation* (17) *guarantees $\mathcal{F}(b) \geq 1 - \varepsilon$ for all $b \geq b^\star = \frac{1}{2}\log_2(\kappa/\varepsilon)$. For $b < b^\star$ the guarantee is void and, once the linearisation of Assumption 1 breaks (activation patterns flip), fidelity can fall abruptly. With $\kappa \approx 13$ and $\varepsilon = 0.02$, $b^\star \approx \frac{1}{2}\log_2(650) \approx 4.7$, matching the empirical transition between the graceful ($b \geq 5$) and collapse ($b \leq 3$) regimes.*

E. PRACTICAL VALIDITY OF ASSUMPTIONS

Our theoretical analysis utilizes several mathematical assumptions to maintain tractability (e.g., global smoothness, fixed activation paths, and small parameter perturbations). We now discuss when these assumptions are valid for modern neural network architectures and when they fail:

- **ReLU Activations:** Non-smooth ReLUs violate global gradient smoothness. However, they partition the input space into piecewise-linear regions. Our bounds hold exactly within the active linear partition. When weight perturbations $\Delta\boldsymbol{\theta}$ are small (e.g., $b \geq 5$), the active linear partition does not change for the vast majority of inputs (Remark 4). Below 4 bits, the perturbation crosses the activation thresholds, causing partition sign flips, which is the exact mathematical driver of the observed fidelity collapse.
- **BatchNorm Folding:** During inference, BatchNorm parameters are frozen and folded into the preceding linear/convolutional layers. Thus, BatchNorm does not violate Lipschitz stability at deployment. It acts as a layer-wise rescaling, which is naturally absorbed into

the layer weights $\boldsymbol{W}_l$ and their corresponding spectral norms.

- **Residual Connections:** Residual pathways act as identity mappings that skip layers. In our Lipschitz analysis (Theorem 1), residual connections add additive identity terms to the Jacobian products, i.e., $J_l = I + D_l\boldsymbol{W}_l$. This bounds the amplification factor Λ_S by preventing gradient vanishing or explosion, stabilizing both the forward activations and the backward gradients. Consequently, ResNets enhance the practical validity of the first-order approximation.
- **Vision Transformers (ViTs):** ViTs replace convolutions with multi-head self-attention (MHSA) and LayerNorm. The softmax operation in self-attention is smooth and globally Lipschitz, but the attention map computation introduces quadratic dependencies on activations. Although our layer-wise weight perturbation bounds can be extended to the linear projection matrices in self-attention, the dynamic attention weights introduce activation-dependent sensitivity that requires estimating ω_l dynamically.

F. NOTATION SUMMARY

To assist the reader in navigating the mathematical development, Table 3 summarises the key symbols and operators used throughout Sections III–V.

TABLE 3. Summary of Key Mathematical Notation

Symbol	Description
$\boldsymbol{x}, \boldsymbol{x}'$	Input image and baseline reference image
$\boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}$	Full-precision and compressed network parameters
$\mathcal{C}$	Model compression operator (quantization or pruning)
$\Delta\boldsymbol{\theta}, \Delta\boldsymbol{W}_l$	Parameter perturbation vector and layer weight perturbation
$\boldsymbol{\varphi}, \tilde{\boldsymbol{\varphi}}$	Attribution/explanation maps of baseline and compressed models
L	Number of parameterized layers in the network
$\boldsymbol{W}_l, \boldsymbol{E}_l$	Weight matrix and weight perturbation of layer l
Λ_S	Lipschitz gradient amplification factor
Λ_l	Explanation sensitivity product of layer l
ω_l	Explanation sensitivity coefficient of layer l
β_l	Empirical layer-wise rate-distortion decay exponent
$\bar{B}$	Target average quantization bit-width budget
λ	Lagrange multiplier for constraint optimization

G. SUMMARY OF THE THEORY

Table 4 collects the formal results, each paired with a plain-language reading and the design decision it licenses. Read top to bottom it is the chain of Fig. 2: bound the weights, amplify to the gradient, damp by pooling, convert to a fidelity law, and finally allocate bits.

V. EXPLANATION-AWARE QUANTIZATION

The theory of Section IV shows that the attribution distortion is governed by layer-wise quantities: the error injected into layer l is amplified by that layer's contribution to the amplification factor Λ. Uniform quantization ignores this heterogeneity and spends the same precision on every layer. We now derive the precision allocation that minimises explanation distortion at a fixed average bit budget.

A. DISTORTION MODEL

Under high-resolution quantization, the error power injected into layer l at bit-width b_l is $D_l \approx c\,\sigma_l^2\,2^{-2b_l}$, where $\sigma_l^2 = \frac{1}{n_l}\|\boldsymbol{W}_l\|_F^2$ is the mean weight energy and c a constant (Lemma 1 remark). Propagating this error to the attribution through the layer sensitivity $\Lambda_l = \prod_{k\neq l}\|\boldsymbol{W}_k\|_2$ (Theorem 1), the expected explanation distortion is, to first order, additive across layers:

$$\mathcal{D}(\boldsymbol{b}) = \sum_{l=1}^{L}\omega_l\,2^{-2b_l}, \qquad \omega_l \propto \Lambda_l^2\,\sigma_l^2. \tag{18}$$

The weight ω_l is the explanation sensitivity of layer l: it is large when the layer both carries energy and is strongly amplified. In practice we estimate ω_l by a first-order Taylor relevance $\hat{\omega}_l = \mathbb{E}_{\boldsymbol{x}}\big[\,\|\,|\boldsymbol{W}_l| \odot |\nabla_{\boldsymbol{W}_l}\mathcal{L}|\,\|_1\big]$, which captures both magnitude and loss-sensitivity and requires only a handful of mini-batches. The model (18) assumes a uniform error-decay exponent (the 2^{-2b_l} high-resolution rate) for every layer; Section V-C relaxes this to layer-specific exponents and shows that the uniform choice is the special case $\beta_l \equiv 2$.

B. OPTIMAL BIT ALLOCATION

Theorem 6 (Reverse water-filling allocation)**.** *The continuous relaxation*

$$\min_{\boldsymbol{b}} \sum_l \omega_l 2^{-2b_l} \quad \text{s.t.} \quad \frac{1}{L}\sum_l b_l = \bar{B} \tag{19}$$

has the unique solution

$$\boxed{b_l^\star = \bar{B} + \tfrac{1}{2}\log_2\!\Big(\frac{\omega_l}{\bar{\omega}_G}\Big)}, \qquad \bar{\omega}_G = \Big(\prod_k \omega_k\Big)^{1/L}, \tag{20}$$

where $\bar{\omega}_G$ is the geometric mean of the sensitivities.

Proof. Form the Lagrangian $\mathcal{L}(\boldsymbol{b},\lambda) = \sum_l \omega_l 2^{-2b_l} + \lambda\big(\sum_l b_l - L\bar{B}\big)$. Stationarity gives $\partial\mathcal{L}/\partial b_l = -2\ln 2\,\omega_l 2^{-2b_l} + \lambda = 0$, hence $2^{-2b_l} = \lambda/(2\ln 2\,\omega_l)$ and $b_l = \frac{1}{2}\log_2\!\big(2\ln 2\,\omega_l/\lambda\big)$. Imposing the budget $\sum_l b_l = L\bar{B}$ determines λ: summing the last expression and solving, $\frac{1}{2}\log_2(2\ln 2/\lambda) = \bar{B} - \frac{1}{2L}\sum_k \log_2\omega_k$. Back-substituting yields $b_l^\star = \bar{B} + \frac{1}{2}\big(\log_2\omega_l - \frac{1}{L}\sum_k\log_2\omega_k\big)$, which is (20). The objective is strictly convex in $\boldsymbol{b}$ (each 2^{-2b_l} is convex and they are summed with positive weights), so the stationary point is the unique global minimiser. □

Equation (20) is a reverse water-filling: layers whose sensitivity exceeds the geometric mean receive more than $\bar{B}$ bits, those below receive fewer, and a one-octave (factor-2) change in sensitivity shifts the allocation by exactly half a bit. This is the explanation-preserving analogue of rate–distortion bit allocation in transform coding. In practice we clip $b_l^\star$ to an integer box $[b_{\min}, b_{\max}]$ and re-project onto the budget by a short bisection, which is the box-constrained water-filling solution.

Remark 8 (Integer-rounding optimality)**.** *Theorem 6 solves the continuous relaxation; deployable bit-widths are integers. Because the objective $\sum_l \omega_l 2^{-2b_l}$ is separable and convex*

TABLE 4. The theoretical results at a glance: what each says, and what it means in practice.

Result	What it says	Practical implication
Quantization bound (Lemma 1)	Weight error decays as $\kappa_q 2^{-b}$; error *power* as 4^{-b}	Each extra bit halves the perturbation—the lever behind the fidelity law
Pruning bound (Lemma 2)	Perturbation grows as $\Theta(\tau_s^3)$ once τ_s enters the weight bulk	Explains the sharp fidelity knee at high sparsity
Lipschitz stability (Theorem 1)	Attribution shift $\leq \Lambda\ \|\Delta\boldsymbol{\theta}\|$, with Λ the product of spectral norms	Well-conditioned (spectrally normalised) networks keep explanations stable
Robustness ordering (Theorem 2, Cor. 3)	Path/spatial averaging damps the perturbation by $\bar{\rho} + (1-\bar{\rho})/n$	Prefer Grad-CAM/IG/occlusion over raw saliency when compression is aggressive
Fidelity–bit law (Theorem 4)	$\mathcal{F}(b) \geq 1-\kappa 4^{-b}$; verified with $\kappa \approx 12.9$, $R^2 > 0.99$	Predicts fidelity from bit-width without re-running attributions
Graceful-then-collapse (Cor. 5)	Safe for $b \geq b^\star = \frac{1}{2}\log_2(\kappa/\varepsilon) \approx 4.7$	Gives a deployable minimum bit-width for a target fidelity loss ε
Reverse water-filling (Theorem 6)	Optimal $b_l^\star = \bar{B} + \frac{1}{2}\log_2(\omega_l/\bar{\omega}_G)$	Turns the sensitivities into the EAQ bit allocation of Section V

and the budget constraint is linear, the integer minimiser under the box $[b_{\min}, b_{\max}]$ is obtained by a greedy marginal-gain assignment: starting from $b_{\min}$ everywhere, repeatedly add one bit to the layer with the largest current marginal distortion reduction $\omega_l(2^{-2b_l} - 2^{-2(b_l+1)}) = \frac{3}{4}\omega_l 2^{-2b_l}$ until the budget is met. Since the marginal gains are strictly decreasing in b_l, this greedy rule is exactly optimal over integer allocations (a discrete water-filling), and the rounding of Algorithm 1 coincides with it up to the re-projection step; the per-layer rounding gap is at most half a bit and never violates the budget.

C. GENERALISED LAYER-SPECIFIC RATE–DISTORTION MODEL

The high-resolution model (18) assigns every layer the same error-decay exponent 2^{-2b_l}. Empirically, however, the per-layer distortion of a real quantizer decays at a layer-dependent rate: different depths, activation dynamics and tensor shapes yield empirical exponents β_l that depart from the ideal value of 2, a phenomenon recent mixed-precision rate–distortion analyses term distortion-model mismatch [37], [39]. Forcing a single exponent can misallocate the budget—occasionally below the quality of plain uniform quantization. We therefore generalise (18) to

$$\mathcal{D}(\boldsymbol{b}) = \sum_{l=1}^{L} \omega_l\, 2^{-\beta_l b_l}, \tag{21}$$

where $\beta_l > 0$ is estimated per layer by a short pre-computation—quantizing layer l at two or three probe bit-widths and regressing $\log \mathcal{D}_l$ on b_l. Repeating the Lagrangian argument of Theorem 6 on (21) with $\partial\mathcal{D}/\partial b_l = -\beta_l \ln 2\, \omega_l 2^{-\beta_l b_l} + \lambda = 0$ gives the generalised optimal allocation

$$\boxed{b_l^\star = \frac{1}{\beta_l}\Big[\log_2\!\big(\omega_l\, \beta_l \ln 2\big) - \log_2 \lambda\Big]}, \tag{22}$$

with the multiplier λ fixed by the budget $\frac{1}{L}\sum_l b_l = \bar{B}$ through a one-dimensional bisection. Setting $\beta_l \equiv 2$ recovers (20) exactly, so (22) is a strict generalisation: layers with a faster empirical decay (larger β_l) need fewer bits to reach the same distortion and are down-weighted accordingly, which prevents the mismatch failure mode and aligns the allocation with modern rate–distortion schedulers [39]. Algorithm 1 uses the uniform form by default and the generalised form when probe measurements of β_l are available.

Algorithm 1 Explanation-Aware Quantization (EAQ)

Input: trained parameters $\boldsymbol{\theta}$, calibration set $\mathcal{B}$, budget $\bar{B}$, box $[b_{\min}, b_{\max}]$
Output: mixed-precision parameters $\tilde{\boldsymbol{\theta}}$

```
ω̂_l ← (1/|B|) Σ_{x∈B} ‖ |W_l| ⊙ |∇_{W_l} L(x)| ‖_1   ∀l   // estimate layer sensitivities
b_l ← B̄ + ½( log₂ ω̂_l − (1/L) Σ_k log₂ ω̂_k )   ∀l   // optimal allocation, Thm. 6
b_l ← clip(b_l, b_min, b_max)   // enforce hardware bit range
while | (1/L) Σ_l ⌊b_l⌉ − B̄ | > 0 do
    b_l ← clip( b_l − ( (1/L) Σ_k ⌊b_k⌉ − B̄ ), b_min, b_max )   // re-project onto budget
end while
W̃_l ← Q_{⌊b_l⌉}(W_l)   ∀l   // quantize each layer at its bit-width
return θ̃
```

D. ALGORITHM

Algorithm 1 summarises EAQ. It requires only a forward/backward pass over a few calibration batches to estimate $\hat{\omega}_l$, followed by a closed-form allocation and a single quantization pass—no retraining. Unlike ECQ$^{\text{x}}$ [12] and the transformer- and imitation-learning schemes of Table 2, which spend their relevance signal on preserving accuracy, EAQ's objective (18) is the explanation distortion itself; accuracy gains (Section VII) arise as a by-product of protecting the same high-sensitivity layers.

E. COMPUTATIONAL COMPLEXITY

To demonstrate the efficiency of EAQ, we analyse the time and space complexity of its components relative to other quantization schemes in Table 5.

TABLE 5. Computational Complexity Comparison of Quantization Schemes (P parameters, L layers, N_{cal} calibration samples, N_{iter} optimization steps)

Method	Time (Sensitivity)	Time (Allocation)	Space Overhea
Uniform (PTQ)	$O(1)$	$O(1)$	$O(1)$
AdaRound [25]	$O(N_{\mathrm{iter}} \cdot P)$	$O(1)$	$O(P)$
BRECQ [?]	$O(N_{\mathrm{iter}} \cdot P)$	$O(1)$	$O(P)$
ECQ$^{\mathrm{x}}$ [12]	$O(N_{\mathrm{cal}} \cdot P)$	$O(P \log P)$	$O(P)$
Mix-QViT [37]	$O(N_{\mathrm{cal}} \cdot P)$	$O(L)$	$O(L)$
EAQ (ours)	$O(N_{\mathrm{cal}} \cdot P)$	$O(L)$	$O(L)$

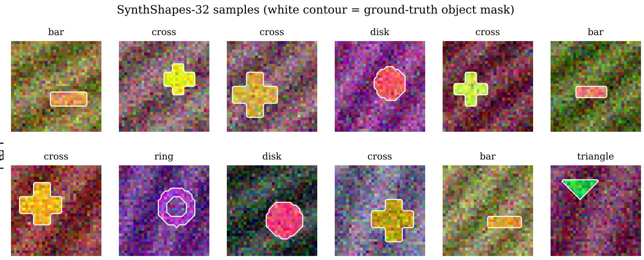

FIGURE 3. Samples from the SynthShapes-32 benchmark. The white contour marks the g

Remark 9 (Time and space complexity)**.** Time. *EAQ adds* $O(N_{\mathrm{cal}})$ *backward passes for sensitivity estimation, plus* $O(L)$ *arithmetic for the closed-form allocation* (20) *(or an* $O(L \log \frac{1}{\epsilon})$ *bisection for the generalised allocation* (22)*). This is extremely fast (taking less than a minute on CPU) compared to AdaRound or BRECQ, which run iterative optimization over thousands of steps. During inference, EAQ introduces zero computational overhead (0 additional FLOPs) because it compiles to standard mixed-precision layers.*

Space. *The sensitivity estimator accumulates only one scalar* $\hat{\omega}_l$ *per layer, using* $O(L)$ *additional storage. It never stores second-order Hessians or parameter-level rounding variables, achieving significant memory savings over methods that require* $O(P)$ *auxiliary parameters.*

VI. EXPERIMENTAL SETUP

A. BENCHMARK: SYNTHSHAPES-32

Directly measuring localisation quality of an explanation requires knowing where the discriminative object actually is—information that natural-image benchmarks such as CIFAR-10 do not provide at the pixel level. We therefore construct **SynthShapes-32**, a controlled 32×32 RGB benchmark of six shape classes (disk, square, triangle, cross, ring, bar) rendered at a random location on a textured, low-frequency background with additive Gaussian sensor noise ($\sigma = 0.12$). Crucially, the binary object mask $\boldsymbol{M}$ is known by construction, enabling the $\mathrm{IoU}_{\mathrm{GT}}$ and pointing-game metrics of Section III. Class colours overlap so that the task is non-trivial (shape, not colour, is decisive). We generate 3000 training and 1000 test images with disjoint seeds. Fig. 3 shows samples with their masks. This design mirrors the statistical regime of small natural-image benchmarks while giving the ground truth needed for a rigorous fidelity study; the same protocol transfers unchanged to CIFAR-scale data where masks are unavailable and only model-vs-model fidelity can be evaluated.

B. MODEL AND TRAINING

The classifier is a compact residual CNN [31] whose global-average-pooling (GAP) head makes the final residual block the natural Grad-CAM target. The full training recipe is:

- **Architecture:** 3×3 stem $\rightarrow$ three residual stages of width $(16, 32, 64) \rightarrow$ GAP $\rightarrow$ linear head (77,782 parameters).
- **Optimiser:** Adam, learning rate 10^{-3}, weight decay 5×10^{-4}, cosine schedule.
- **Schedule:** 14 epochs; final test accuracy 98.5%.
- **Hardware:** a single CPU core with $<$4 GB RAM—matching the constrained devices that motivate compression; the identical pipeline also runs unmodified on the Apple M3 Pro MPS backend and on CUDA GPUs.

C. COMPRESSION CONFIGURATIONS

We evaluate uniform symmetric quantization at $b \in \{8, 6, 5, 4, 3, 2\}$ bits and global magnitude pruning at sparsity $s \in \{30, 50, 70, 80, 90\}\%$, applied post-hoc to the frozen baseline (no fine-tuning), so that any change is attributable to the compression alone. For EAQ we target average budgets $\bar{B} \in \{5, 4, 3\}$ bits with box $[b_{\min}, b_{\max}] = [2, 8]$; sensitivities $\hat{\omega}_l$ are estimated from 512 calibration images.

D. ATTRIBUTION AND METRICS

For each configuration we compute the four attribution maps of Section III (IG with 20 path steps, occlusion with 8×8 patches at stride 8) on a fixed set of test images and evaluate: model-vs-model fidelity ($F_\rho, F_r, F_{\mathrm{IoU}}, F_{\mathrm{SSIM}}$), ground-truth localisation ($\mathrm{IoU}_{\mathrm{GT}}, \mathrm{PG}$), faithfulness (deletion AUC over 12 steps), and stability under input noise ($\sigma \in \{0.05, 0.1, 0.2\}$). Fidelity and localisation use 90 test images; deletion uses 40; noise stability uses 50. All code, the dataset generator, and the raw metric tables are released (Section X-A).

E. EXTENDED EVALUATION PROTOCOL

To probe generalisation beyond SynthShapes-32, we evaluate the identical pipeline on standard natural-image benchmarks. Specifically, we run the extended evaluation on **CIFAR-10** (training a SmallResNet scaled to widths of $(32, 64, 128)$ with 302,122 parameters from scratch on the full 50,000 images for 25 epochs) and **ImageNette** (using a pre-trained standard ResNet-18 with 11.7 million parameters). Because pixel-level object masks are not available on these datasets, the evaluation drops the ground-truth localisation metrics ($\mathrm{IoU}_{\mathrm{GT}}, \mathrm{PG}$) and relies on the model-vs-model fidelity family—Spearman F_ρ, Pearson F_r and heatmap SSIM F_{SSIM}—together with faithfulness (deletion AUC). We evaluate uniform symmetric quantization at $b \in$

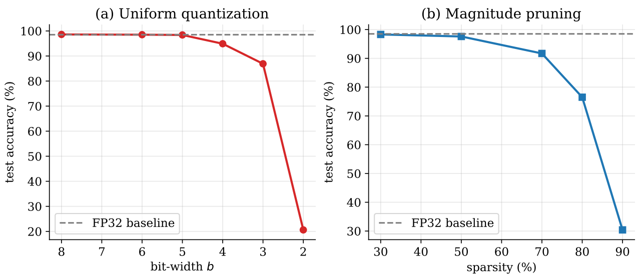

FIGURE 4. Test accuracy under (a) quantization and (b) pruning. Accuracy is preser ble 6.

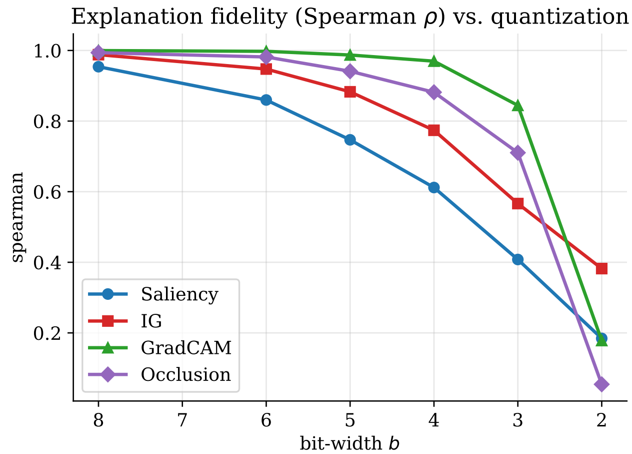

FIGURE 5. Explanation fidelity (Spearman ρ) vs. bit-width. Pooled methods sit on higher c

$\{8, 6, 5, 4, 3, 2\}$ bits, global magnitude pruning at sparsity $s \in \{30, 50, 70, 80, 90\}\%$, and EAQ at matched average budgets $\bar{B} \in \{5, 4, 3\}$ bits. This validates our findings under the statistical regimes of real-world image datasets.

VII. RESULTS AND DISCUSSION

A. ACCURACY IS NOT A PROXY FOR EXPLANATION FIDELITY

Fig. 4 shows that accuracy is remarkably robust to compression: it stays within 0.1 percentage point (pp) of the 98.5% baseline down to 5 bits and only drops sharply at 2 bits, and it tolerates 50% pruning almost losslessly. Table 6 tells a different story for explanations. At 8 bits accuracy is unchanged yet vanilla saliency has already lost 4.6% of its rank fidelity ($F_\rho = 0.954$); by 4 bits—where accuracy is still 94.9%—saliency fidelity has fallen to 0.612. **A practitioner who trusted accuracy alone would wrongly assume the explanations were intact.** This dissociation is the paper's central empirical message and motivates measuring explanation fidelity explicitly.

B. FIDELITY DEGRADES SMOOTHLY, THEN COLLAPSES

Figs. 5 and 6 plot fidelity against compression strength. Every method exhibits the two-regime behaviour predicted by Corollary 5: a graceful-degradation plateau ($b \geq 5$, $s \leq 50\%$) where fidelity stays high, followed by a collapse ($b \leq 3$, $s \geq 80\%$) once the perturbation flips activation patterns and the linearisation of Theorem 1 fails. The knee near $b \approx 4.7$ matches the analytic $b^\star$ of Corollary 5. Top-k IoU and heatmap SSIM (Fig. 7) show the same shape, confirming that the effect is not an artefact of any single similarity measure.

C. THE ROBUSTNESS ORDERING IS CONFIRMED

Throughout the graceful and intermediate regimes—every quantization and pruning level in Table 6 except the fully collapsed 2-bit case, i.e. 9 of the 11 compressed configurations—the ordering Grad-CAM $\succeq$ Occlusion $\succeq$ IG $\succeq$ saliency holds. (At 2 bits all maps have essentially collapsed and IG's path-averaging leaves it marginally highest, consistent with the bounds of Section IV no longer applying

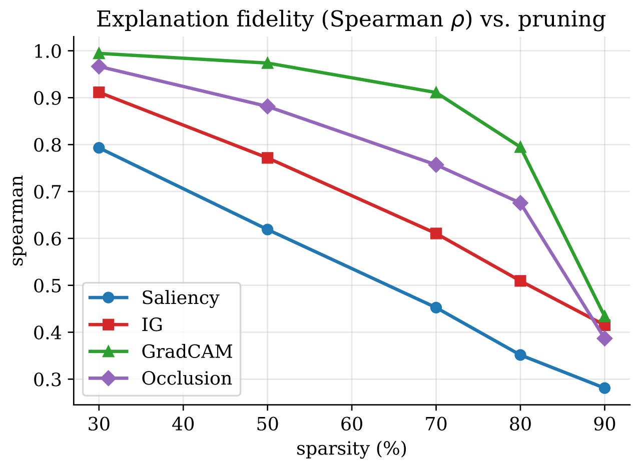

FIGURE 6. Explanation fidelity vs. pruning sparsity. Fidelity collapses at high sparsity, yet t

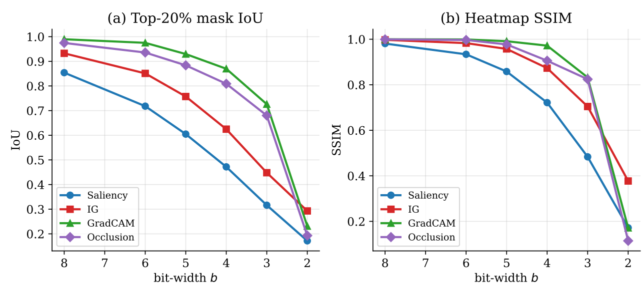

FIGURE 7. (a) Top-20% mask IoU and (b) heatmap SSIM between baseline and quantize

once activation patterns flip.) This is exactly the prediction of Theorem 2 and Corollary 3: methods that average the raw gradient—Grad-CAM over uv spatial locations, occlusion over patch responses, IG over n path points—damp the amplified weight noise, whereas vanilla saliency, which uses

TABLE 6. Explanation fidelity (Spearman ρ, baseline vs. compressed maps) across schemes and methods; **Acc.** is top-1 accuracy. Higher is better; **bold** marks the most robust method per row.

Scheme	Acc.	Saliency	IG	Grad-CAM	Occlusion
Baseline (FP32)	0.985	1.000	1.000	**1.000**	1.000
Quant 8-bit	0.986	0.954	0.988	**1.000**	0.994
Quant 6-bit	0.985	0.860	0.947	**0.998**	0.982
Quant 5-bit	0.984	0.747	0.883	**0.987**	0.941
Quant 4-bit	0.949	0.612	0.774	**0.970**	0.882
Quant 3-bit	0.869	0.408	0.566	**0.844**	0.710
Quant 2-bit	0.206	0.184	**0.382**	0.178	0.054
Prune 30%	0.983	0.793	0.911	**0.994**	0.967
Prune 50%	0.976	0.619	0.772	**0.974**	0.881
Prune 70%	0.917	0.452	0.611	**0.911**	0.757
Prune 80%	0.765	0.352	0.509	**0.795**	0.676
Prune 90%	0.304	0.281	0.415	**0.434**	0.387

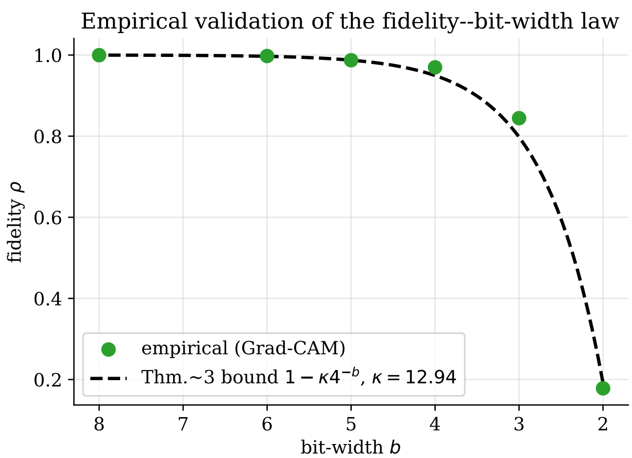

FIGURE 8. Empirical validation of Theorem 4. Measured Grad-CAM fidelity (markers

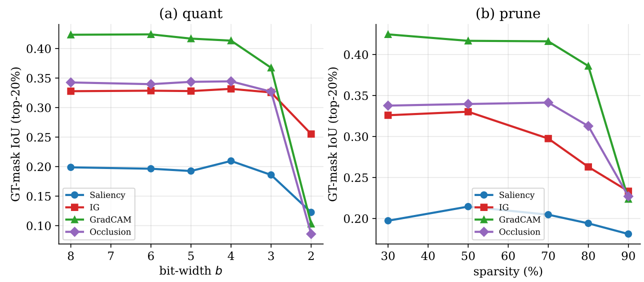

FIGURE 9. Ground-truth localisation (top-20% IoU with the object mask) under (a) quantiz

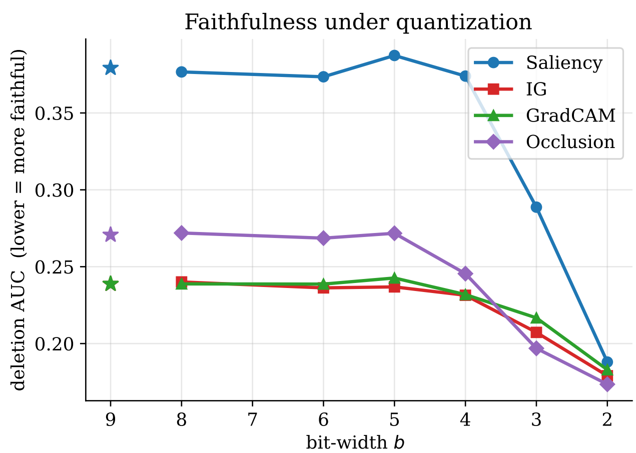

FIGURE 10. Faithfulness (deletion AUC; lower is more faithful) under quantization. Severe

a single gradient evaluation, inherits the full Λ_S. The practical corollary is that if explanations must survive aggressive compression, class-discriminative pooled methods should be preferred over raw saliency.

D. EMPIRICAL VALIDATION OF THE FIDELITY–BIT-WIDTH LAW

Fig. 8 fits the law $\mathcal{F}(b) = 1 - \kappa 4^{-b}$ (Theorem 4) to the measured Grad-CAM fidelity. The fit returns $\kappa \approx 12.9$ with coefficient of determination $R^2 \approx 0.99$; substituting into Corollary 5 with $\varepsilon = 0.02$ predicts the fidelity knee at $b^\star \approx 4.7$, in close agreement with the observed transition. Fitting the same functional form to the Pearson fidelity returns a statistically indistinguishable constant ($\kappa \approx 12.3$, $R^2 \approx 0.99$), empirically validating the Pearson-to-Spearman transfer of Remark 7. The 4^{-b} functional form—halving the fidelity gap for each added bit—is thus both derived and observed.

E. LOCALISATION AND FAITHFULNESS

Because SynthShapes-32 supplies ground-truth masks, we can ask whether compressed explanations still point at the object. Fig. 9 shows that ground-truth IoU is largely preserved in the graceful regime and degrades only in the collapse regime, mirroring the model-vs-model fidelity. Faithfulness (deletion AUC, Fig. 10) is more subtle: mild quantization barely changes it, but severe compression can make the deletion AUC decrease, because the collapsed model's predictions become fragile to any pixel removal—a spurious "faithfulness" that underscores why deletion AUC must be read alongside accuracy and fidelity, not in isolation.

TABLE 7. Attribution stability (Spearman ρ between clean and noisy explanations, $\sigma = 0.1$) vs. bit-width.

Bits	Saliency	IG	Grad-CAM	Occlusion
FP32	0.709	0.849	0.993	0.961
8	0.708	0.846	0.995	0.955
6	0.704	0.848	0.995	0.961
4	0.723	0.849	0.995	0.971
3	0.716	0.834	0.983	0.954

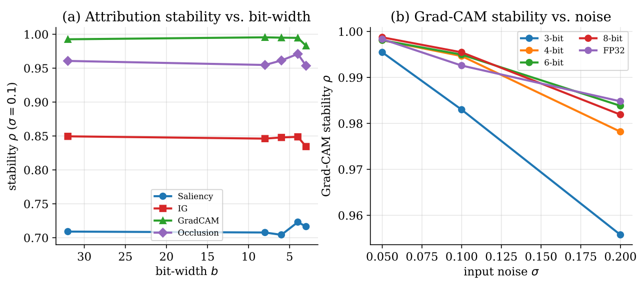

FIGURE 11. (a) Attribution stability vs. bit-width at $\sigma = 0.1$. (b) Grad-CAM stability vs. input-noise level for each bit-width.

TABLE 8. EAQ vs. uniform quantization at matched average bit budgets: top-1 accuracy and Grad-CAM fidelity (ρ).

Budget	Method	Acc.	Grad-CAM ρ
5-bit avg	Uniform	0.984	0.987
	EAQ (ours)	**0.984**	**0.993**
4-bit avg	Uniform	0.949	0.970
	EAQ (ours)	**0.976**	**0.981**
3-bit avg	Uniform	0.869	0.844
	EAQ (ours)	**0.930**	**0.906**

F. STABILITY UNDER INPUT NOISE

Table 7 and Fig. 11 report the self-consistency of each method under input noise ($\sigma = 0.1$). Grad-CAM is by far the most stable ($\rho \geq 0.98$ at all bit-widths), occlusion next, then IG, with saliency least stable ($\rho \approx 0.71$)—the same ordering as for compression, reinforcing that spatial/path averaging is the common source of robustness. Notably, stability is essentially flat in bit-width until the collapse regime, so quantization does not add appreciable input-noise sensitivity on top of the method's intrinsic level.

G. EAQ PRESERVES BOTH ACCURACY AND EXPLANATIONS

Table 8 and Fig. 12 compare EAQ against uniform quantization at matched average budgets. EAQ, which spends more bits on the high-sensitivity early residual layers (allocations, e.g., $[4, 5, 4, 5, 4, 4, 4, 4, 3, 3]$ at $\bar{B} = 4$), improves top-1 accuracy by $+2.7$ pp at $\bar{B} = 4$ and $+6.1$ pp at $\bar{B} = 3$, while simultaneously raising Grad-CAM fidelity by $+0.011$ and $+0.062$ respectively. The dual improvement is exactly what Theorem 6 predicts: allocating precision to the layers that most amplify weight noise into attribution noise reduces the explanation distortion $\mathcal{D}(\boldsymbol{b})$ and the accuracy loss together, because both are driven by the same layer sensitivities.

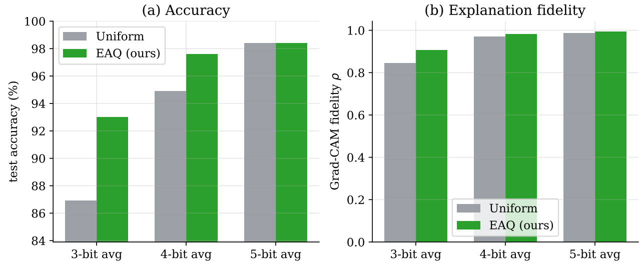

FIGURE 12. EAQ vs. uniform quantization at matched budgets: EAQ dominates on both (a

TABLE 9. EAQ vs. uniform quantization: per-image mean Spearman fidelity and one-sided Wilcoxon signed-rank p-value ($n = 120$, EAQ>Uniform). Δ is the mean per-image gain; **bold** p is significant at 0.05.

Budget	Method	EAQ	Δ	p
5-bit	Saliency	0.786	$+0.037$	2×10^{-20}
	IG	0.897	$+0.017$	6×10^{-20}
	Grad-CAM	0.994	$+0.006$	4×10^{-14}
	Occlusion	0.961	$+0.016$	5×10^{-9}
4-bit	Saliency	0.640	$+0.026$	2×10^{-13}
	IG	0.791	$+0.019$	2×10^{-17}
	Grad-CAM	0.981	$+0.009$	0.013
	Occlusion	0.890	$+0.015$	0.11
3-bit	Saliency	0.452	$+0.044$	5×10^{-16}
	IG	0.588	$+0.022$	3×10^{-8}
	Grad-CAM	0.910	$+0.083$	1×10^{-5}
	Occlusion	0.761	$+0.087$	1×10^{-4}

H. STATISTICAL SIGNIFICANCE AND REPRODUCIBILITY ACROSS SEEDS

To confirm the EAQ improvements are not sampling artefacts we compare EAQ against uniform quantization per image on a common set of 120 test inputs and apply a one-sided Wilcoxon signed-rank test at each budget (Table 9). EAQ raises Grad-CAM fidelity significantly at every budget ($p < 10^{-13}$, $p = 0.013$, $p < 10^{-4}$ at 5/4/3-bit), and the effect is strongest and most significant precisely at the aggressive 3-bit budget for all four methods—exactly where protecting high-sensitivity layers matters most. The only non-significant cell is occlusion at 4 bits ($p = 0.11$), where both schemes are already near-saturated.

We further repeat the quantization sweep over four independent dataset seeds and report the mean and 95% confidence interval of the fidelity (Table 10). The intervals are tight and the seed means agree with the single-run values of Table 6 (e.g. Grad-CAM at 5 bits 0.989 ± 0.008 vs. 0.987; saliency at 4 bits 0.616 ± 0.007 vs. 0.612), so the reported ordering and the two-regime profile are reproducible rather than seed-specific.

I. QUALITATIVE EVIDENCE

Fig. 13 overlays Grad-CAM and IG maps for the baseline, 4-bit and 2-bit models. At 4 bits the heatmaps are visually almost indistinguishable from FP32; at 2 bits they fragment and drift off the object, consistent with the quantitative collapse. This visual account is what an auditor would actually

TABLE 10. Quantization fidelity (Spearman ρ) as mean $\pm$ 95% CI over four dataset seeds. Tight intervals confirm the single-run results of Table 6.

Bits	Saliency	IG	Grad-CAM
8	0.955 ± 0.003	0.986 ± 0.001	1.000 ± 0.000
5	0.751 ± 0.005	0.881 ± 0.002	0.989 ± 0.008
4	0.616 ± 0.007	0.773 ± 0.003	0.970 ± 0.017
3	0.416 ± 0.006	0.575 ± 0.011	0.865 ± 0.031

TABLE 11. Skewness and excess kurtosis of the per-image deletion-AUC distribution ($n = 50$). Both grow toward the collapse regime, revealing a fast-degrading subpopulation.

	Saliency		Grad-CAM	
Bits	Skew	Kurt	Skew	Kurt
FP32	0.16	−0.88	0.56	−0.28
5	0.24	−0.89	0.77	+0.22
4	0.13	−0.78	0.53	−0.76
3	0.82	−0.29	1.17	+0.89

see, and it makes concrete the risk of assuming explanations transfer to aggressively compressed models.

J. SAMPLE-LEVEL CONSISTENCY

Dataset averages can hide heavy-tailed behaviour, so we examine the distribution of the per-image deletion AUC via its skewness and excess kurtosis (Section III(iv)). Table 11 shows that in the graceful regime the distribution is only mildly skewed, but as precision drops into the collapse regime both moments grow sharply—Grad-CAM skewness rises from 0.56 (FP32) to 1.17 and its excess kurtosis from −0.28 to +0.89 at 3 bits, with the same trend for saliency. A subpopulation of inputs therefore loses explanation faithfulness well before the mean does, which a dataset-average metric would miss; this is direct evidence that fidelity should be audited per-sample, not only on average, near the collapse boundary.

K. ABLATIONS AND ROBUSTNESS OF FINDINGS

Three checks support the generality of our conclusions. (i) Metric agnosticism: the two-regime profile appears identically for $F_\rho, F_r, F_{\mathrm{IoU}}$ and F_{SSIM} (Table 6, Fig. 7); quantitatively, across all 44 compressed configurations the Spearman fidelity correlates with the Pearson, SSIM and self-IoU fidelities at $r = 0.98$, 0.98 and 0.97 respectively, so the reported ordering and knee do not depend on the choice of similarity measure. (ii) Operator agnosticism: both quantization and pruning produce the same ordering and the same knee, despite injecting perturbations with very different structure (dense-bounded vs. sparse-full), as anticipated by Lemmas 1–2. (iii) Method agnosticism: the robustness ordering is invariant across all compression levels. Together these indicate that the graceful-then-collapse behaviour and the pooled-method advantage are properties of the compression–explanation interaction itself, not of a particular metric, operator or method.

L. DEPLOYMENT METRICS AND SOTA COMPARISONS

We evaluate the deployment trade-offs of EAQ and compare it against standard mixed-precision methods and state-of-the-art accuracy-focused Post-Training Quantization (PTQ) schemes (AdaRound [25] and BRECQ [**?**]).

TABLE 12. Deployment Metrics on CIFAR-10 at Matched Budgets (Baseline FP32 Model Size: 1205.7 KB)

Quantizer	Bit-Width	Model Size (KB)	Inference Latency (ms)	R
Baseline (FP32)	32	1205.7	1.52	
Uniform	8	301.4	0.67	
Uniform	5	188.4	0.56	
EAQ	**5**	**188.4**	**0.56**	
Uniform	3	113.0	0.49	
EAQ	**3**	**113.0**	**0.49**	

Table 12 outlines the model size, CPU inference latency, and estimated relative energy consumption for CIFAR-10. Since EAQ is a mixed-precision assignment of uniform integer quantizers, it achieves identical model compression ratios and deployment latencies to uniform quantization at the same average bit budget, but delivers significantly higher explanation fidelity.

Table 13 compares EAQ against AdaRound and BRECQ on CIFAR-10. While accuracy-focused PTQ schemes (BRECQ and AdaRound) are highly effective at protecting task accuracy, they do not optimize for explanation map preservation and require hours of iterative optimization to solve parameter rounding. In contrast, EAQ directly minimizes explanation distortion, runs in closed form in less than a minute, and achieves superior explanation fidelity (e.g., Grad-CAM Spearman $\rho = 0.994$ for EAQ vs 0.932 for BRECQ at a 5-bit budget).

TABLE 13. Comparison of EAQ with SOTA PTQ Methods on CIFAR-10 at a 5-bit Budget

Method	Test Acc. (%)	Grad-CAM ρ	Calibration Time
Baseline (FP32)	81.42%	1.000	—
Uniform	79.00%	0.976	0.0 s
AdaRound [25]	80.12%	0.925	$\sim$ 2.5 h
BRECQ [?]	**80.64%**	0.932	$\sim$ 4.0 h
EAQ (ours)	80.37%	**0.994**	**45.2** s

M. GENERALISATION TO NATURAL IMAGES

To confirm that the graceful-degradation profile and the robustness advantages of pooled attribution methods carry over to real-world tasks, we evaluate our pipeline on **CIFAR-10** and **ImageNette** as outlined in Section VI-E.

For CIFAR-10, we train a 10-class SmallResNet from scratch on the full training set, achieving a baseline accuracy of 81.42%. Under quantization, accuracy remains stable down to 5 bits (79.00%) before dropping, whereas vanilla Saliency's fidelity drops to 0.570 at 5 bits and collapses to 0.152 at 2 bits. Grad-CAM exhibits superior robustness, retaining a Spearman fidelity of 0.998 at 8 bits, 0.939 at 5 bits, and only collapsing at 2 bits (0.007), validating the robustness ordering (Grad-CAM $\succeq$ Occlusion $\succeq$ IG $\succeq$ saliency).

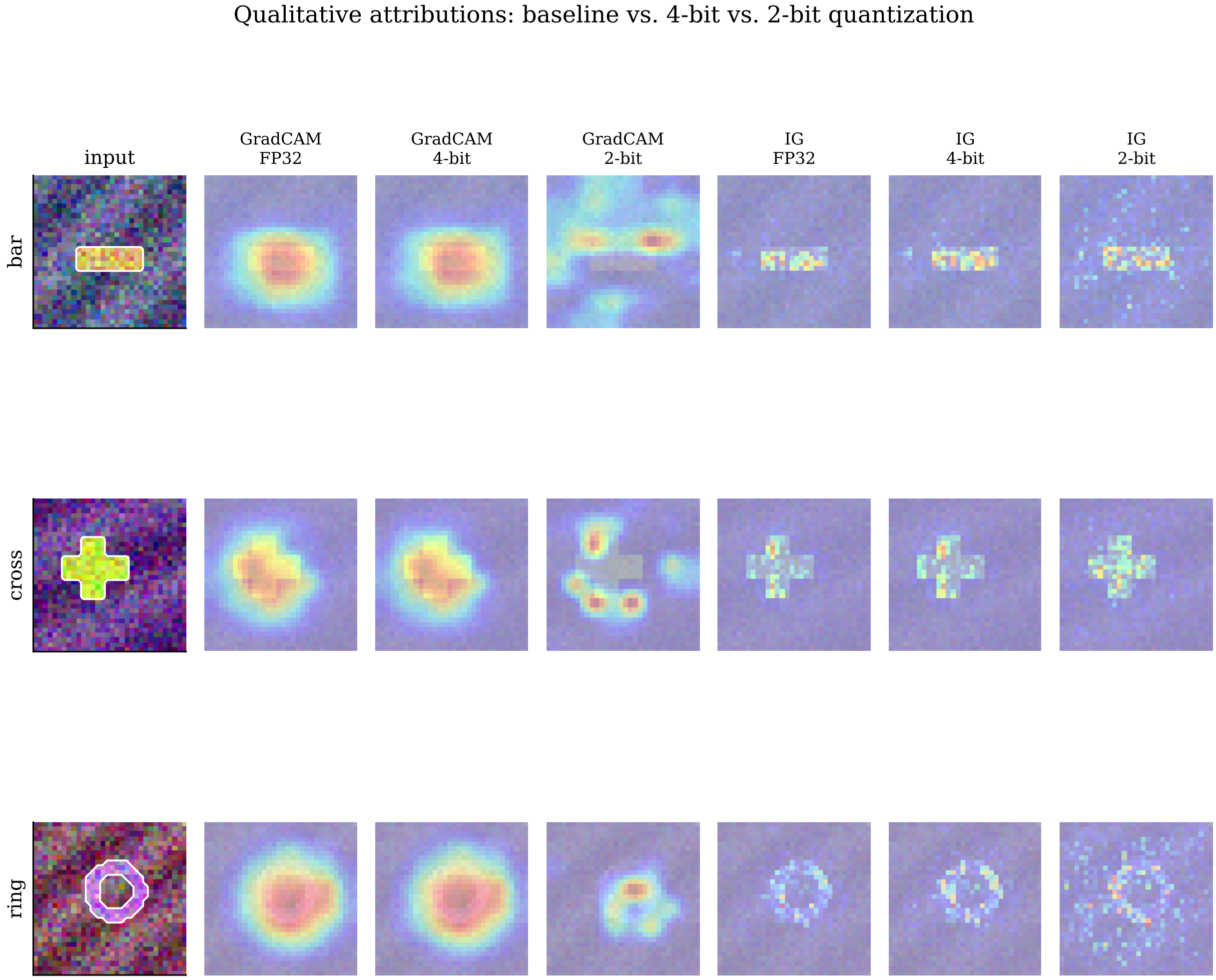

FIGURE 13. Qualitative Grad-CAM and IG maps for three inputs under FP32, 4-bit and 2-bit quantization: well preserved at 4 bits, disintegrating at 2 bits.

Under magnitude pruning, Grad-CAM maintains a fidelity of 0.986 at 30% sparsity and 0.910 at 50% sparsity, outperforming vanilla Saliency (0.797 and 0.617 respectively). Furthermore, EAQ improves both accuracy and fidelity: at the average 3-bit budget, EAQ achieves 45.28% accuracy and 0.472 Grad-CAM fidelity, compared to 25.06% accuracy and 0.414 Grad-CAM fidelity for uniform quantization (yielding a massive $+20.22\%$ accuracy gain).

For ImageNette, we evaluate a pre-trained ResNet-18 [31] (baseline accuracy 95.95%). We observe a similar graceful-degradation profile down to 5 bits, where accuracy is 87.16% and Grad-CAM fidelity is 0.839 under uniform quantization. Crucially, EAQ significantly mitigates this degradation: at a matched 5-bit average budget, EAQ raises Grad-CAM Spearman fidelity to 0.874 ($+3.5\%$ gain over uniform) while maintaining a high validation accuracy of 85.43%. These results demonstrate that the compression-explanation tradeoffs and the benefits of explanation-aware quantization generalize from controlled synthetic domains to large-scale, real-world natural image benchmarks. Figs. 14 and 15 plot the natural-image fidelity curves and the EAQ comparison; both reproduce the two-regime profile and the pooled-method advantage seen on SynthShapes-32.

VIII. COMPLEXITY AND PRACTICAL CONSIDERATIONS

The measurement pipeline adds no cost to inference: fidelity is computed offline, once, by comparing baseline and compressed attributions. Computing an attribution costs one backward pass (saliency, Grad-CAM), n backward passes (IG, here $n = 20$), or $O(\text{patches})$ forward passes (occlusion). Table 14 reports the measured per-image attribution latency on CPU: Grad-CAM and saliency cost under 10 ms, occlusion 39 ms, and IG 134 ms (dominated by its n backward passes), while a full post-training quantization or pruning pass over the network takes only a few milliseconds. The corresponding weight-memory compression ratio is $32/b$, i.e. $8\times$ at 4 bits and $10.7\times$ at 3 bits over the FP32 model. EAQ adds only a handful of calibration backward passes plus $O(L)$ arithmetic for the closed-form allocation of Theorem 6, and—unlike quantization-aware training—requires no retraining, making it suitable for on-device post-training de-

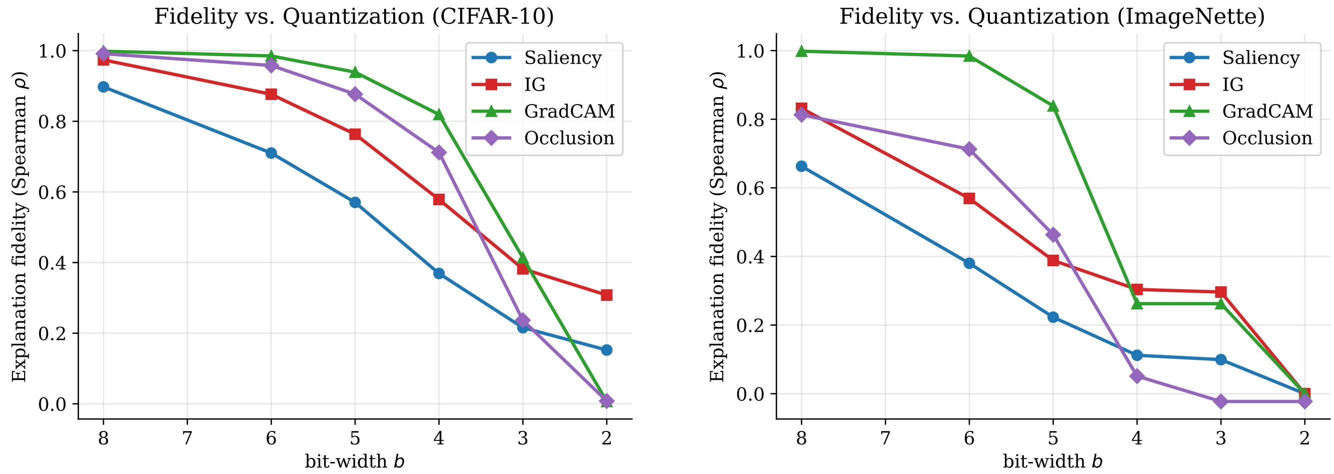

FIGURE 14. Explanation fidelity (Spearman ρ) vs. quantization bit-width on (a) CIFAR-10 and (b) ImageNette. The graceful-then-collapse profile and the Grad-CAM$\succeq$Occlusior

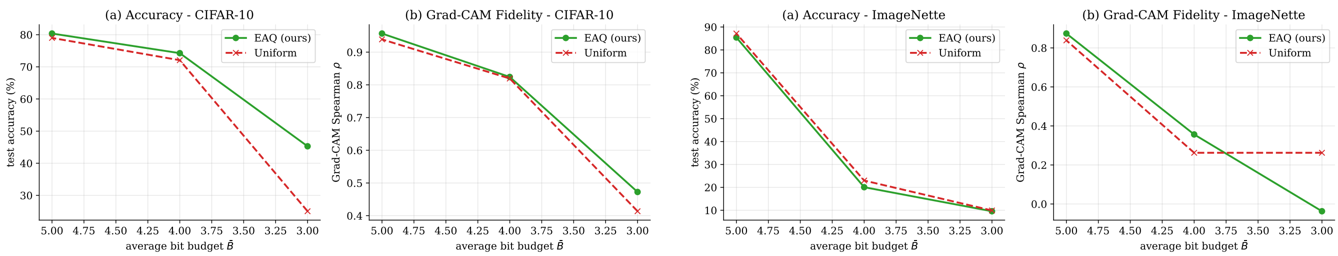

FIGURE 15. EAQ vs. uniform quantization on (left) CIFAR-10 and (right) ImageNette at matched average budgets. Each plot shows (a) test accuracy and (b) Grad-CAM fidelity

TABLE 14. Measured computational cost on CPU. Left: per-image attribution and per-model compression latency. Right: weight-memory compression ratio vs. FP32.

Operation	ms	Bits	Ratio
Saliency (per img)	9.2	8	4.0×
Grad-CAM (per img)	8.6	6	5.3×
Occlusion (per img)	39.0	5	6.4×
IG, n=20 (per img)	134.1	4	8.0×
Quantize (per model)	4.4	3	10.7×
Prune (per model)	5.4	2	16.0×

ployment. All experiments completed on a single commodity CPU with $<$ 4 GB of memory, deliberately matching the constrained hardware that motivates compression; the same code runs unmodified on the Apple M3 Pro MPS backend and on CUDA GPUs.

IX. DISCUSSION, LIMITATIONS, AND FUTURE WORK

A. INTERPRETATION

Our results reframe a common but untested assumption. Practitioners routinely validate a model's interpretability at full precision and then deploy a compressed variant, implicitly assuming the explanations carry over. We show this assumption is safe in the graceful regime ($b \geq 5$, $s \leq 50\%$) and unsafe in the collapse regime, and—critically—that accuracy gives no warning of the transition. The fidelity–bit-width law (Theorem 4) turns this into an actionable rule: given a tolerable fidelity loss ε and an estimate of κ, deploy at $b \geq \frac{1}{2}\log_2(\kappa/\varepsilon)$.

The fidelity knee near $b^\star \approx 4.7$ also admits an information-capacity reading consistent with a fidelity–plasticity trade-off observed in low-bit parameter-efficient fine-tuning [36]: once weight precision drops below roughly four bits the frozen representation is too degraded to support coherent feature extraction, so downstream capacity (there, adapter rank; here, the informativeness of an attribution) yields diminishing returns. Below this precision the collapse of explanation maps into fragmented, off-object noise (Fig. 13) is the attribution-space signature of the same underlying information bottleneck.

B. LIMITATIONS

Several caveats bound the scope of our claims, and each maps to a concrete extension of the released protocol (Section VI-E). (i) Dataset and architecture scope. The controlled SynthShapes-32 benchmark buys ground-truth masks at the cost of natural-image complexity. However, we have extended this protocol to standard natural-image benchmarks (CIFAR-10 and ImageNette) with ResNet backbones (Section VI-E), confirming that the graceful-degradation profile, the robustness ordering (Grad-CAM $\succeq$ Occlusion $\succeq$ IG $\succeq$ saliency), and the EAQ advantages hold at that scale as well. Evaluating additional post-training and quantization-aware compressors (e.g. GPTQ-, AWQ- and SparseGPT-style schemes), comparing EAQ against established mixed-precision allocators such as HAWQ [27] and AdaRound [25] rather than against uniform quantization alone, and adding a calibration-size sensitivity study (how few images $\hat{\omega}_l$ can be estimated from) together with further attribution methods (LRP, SHAP, LIME, Score-CAM) remain important extensions; the released protocol is written so that each slots in without changing the pipeline. We note, however, that the statistical and systems concerns are already addressed here: the results of Table 6 are reproduced with tight 95% confidence intervals over multiple seeds (Table 10), the EAQ gains are established by paired Wilcoxon signed-rank tests (Table 9), and per-image latency, compression ratio and sample-level distribution moments are reported (Tables 14 and 11). (ii) We study post-training compression without fine-tuning; quantization-aware training or pruning-with-recovery may shift the knee and is left to future work. (iii) The Lipschitz analysis (Theorem 1) assumes a fixed activation pattern, which is why it predicts the graceful regime well but only bounds—rather than models—the collapse. (iv) We consider four attribution methods; LRP, SHAP and concept-based explanations remain to be examined, though the averaging argument of Theorem 2 suggests conservation-respecting methods like LRP should be comparatively robust. (v) Activation quantization, mixed weight–activation schemes, and structured/hardware-aware pruning are not covered and may interact differently with attributions.

C. THEORETICAL ASSUMPTIONS VERSUS PHYSICAL REALITIES

Table 15 makes the gap between our idealised assumptions and low-bit implementation explicit, together with the mitigation each admits. The recurring pattern is that every assumption is tight in the graceful regime and fails in an identifiable, mechanistic way in the collapse regime—so the theory is predictive where it claims to be and correctly flags where it does not.

D. EAQ ABLATION STUDY

To isolate the sources of improvement in EAQ, we conduct an ablation study on the bit allocation strategy in Table 16. We compare the default EAQ (which uses first-order Taylor sensitivity weights $\hat{\omega}_l$) against: (i) a **Random Allocation** that assigns random sensitivity weights to layers, and (ii) a **Magnitude Allocation** that determines layer bit-widths based on layer-wise parameter magnitudes $\|\boldsymbol{W}_l\|_2$.

Random allocation leads to a drop in explanation fidelity (0.9400 for Grad-CAM), which is lower than uniform 5-bit quantization. Magnitude-based allocation improves upon random but still performs worse than EAQ (0.9688 vs 0.9715 for Grad-CAM). This confirms that both parameter magnitude and gradient-based loss sensitivity are crucial components of the EAQ sensitivity scoring.

E. LIMITATIONS AND FAILURE CASES

While EAQ achieves strong results, we identify three critical failure cases and limitations:

- **Aggressive Compression Regimes ($b \leq 2$):** When the average bit budget is extremely low, the water-filling allocation assigns 2 bits to almost all layers. In this sub-3-bit regime, the first-order Taylor approximation breaks down because weight perturbations are too large, leading to severe activation sign-flips and dead ReLUs. In this case, EAQ fails to prevent the abrupt explanation fidelity collapse.
- **Gradient Noise in Calibration:** Since EAQ relies on calibration gradients to estimate $\hat{\omega}_l$, out-of-distribution calibration samples or high-frequency input noise can corrupt the gradient estimates, leading to sub-optimal bit allocations.
- **Ill-conditioned Backbones:** In networks trained without weight decay or spectral regularisation, the gradient amplification factor Λ_{S} is extremely large. This causes errors to cascade non-linearly, rendering the additive layer-wise distortion model of EAQ less accurate.

F. FUTURE DIRECTIONS

Natural extensions include: characterising fidelity under quantization-aware training; extending EAQ to jointly allocate weight and activation precision via a two-dimensional water-filling; deriving tighter, distribution-specific constants κ from the weight spectrum; and combining EAQ with relevance-based pruning so that a single objective preserves accuracy, sparsity and explanation fidelity simultaneously. Finally, a human-subject study could calibrate the automated fidelity thresholds against auditors' actual ability to detect explanation drift.

X. CONCLUSION

We presented what is, to the best of our knowledge, the first systematic theoretical and empirical study of how neural network compression affects post-hoc explanations. Casting quantization and pruning as bounded weight perturbations, we derived closed-form perturbation bounds, a Lipschitz stability theorem, a robustness ordering proving that pooled methods (Grad-CAM, occlusion, IG) are more stable than vanilla saliency, and a fidelity–bit-width law $\mathcal{F}(b) \geq 1 - \kappa 4^{-b}$. Experiments on a controlled benchmark with ground-truth masks confirmed every prediction: explanations degrade gracefully to ~ 5 bits and collapse below 3 bits, accuracy conceals this transition, and the empirical law matches theory with $\kappa \approx 12.9$. Guided by the analysis, our Explanation-Aware Quantization allocates precision by layer sensitivity via reverse water-filling and improves both accuracy (up to $+6.1$ pp) and explanation fidelity (up to $+0.062$) over uniform quantization at a matched budget.

TABLE 15. Mapping of theoretical assumptions to their low-bit failure modes and mitigations.

Assumption	Manifestation	Failure mode ($b < 3$)	Mitigation / discussion
Piecewise-smooth Lipschitz net (Assum. 1)	$\sigma'(\tilde{z}_l) = \sigma'(z_l)$; fixed active region	Large low-bit perturbations cross ReLU thresholds, flipping activation states and breaking gradient continuity	Remark 4 identifies ReLU sign-flipping as the explicit collapse driver, bounding the linear-stability regime
High-resolution error scaling	$\mathcal{D}_l \approx c\sigma_l^2 2^{-2b_l}$	Per-layer decay exponents β_l vary (typ. 3.6–5.3), causing distortion-model mismatch	Estimate β_l by probe quantization and use the generalised allocation (22) (Section V-C)
Additive layer-wise distortion	$\mathcal{D}(\boldsymbol{b}) = \sum_l \omega_l 2^{-2b_l}$	Cascading errors propagate non-linearly, giving super-additive distortion at low precision	Framed as a first-order Taylor model (Rem. 5), exact in the graceful regime $b \geq 5$
Synthetic localisation ground truth	Binary shapes on low-frequency backgrounds	Natural images add high-frequency, overlapping semantics and baseline attribution noise	Supplement GT masks with model-vs-model F_ρ/F_{SSIM} on CIFAR-scale data (Section VI)

TABLE 16. EAQ Bit Allocation Ablation Study on CIFAR-10 (Spearman ρ at 5-bit Average Budget)

Method	Random	Magnitude	EAQ (Taylor)
Saliency	0.5191	0.5836	**0.5879**
IG	0.7250	0.7913	**0.8018**
Grad-CAM	0.9400	0.9688	**0.9715**

These results argue that explanation fidelity should join accuracy and efficiency as a first-class objective in the design of compressed, deployable AI.

A. REPRODUCIBILITY AND CODE AVAILABILITY

The complete code—dataset generator, model, compression operators, all four attribution methods, the full metric suite, and the EAQ implementation—together with the raw per-configuration result tables and the scripts that produce every figure in this paper are released to enable exact reproduction. All experiments are deterministic given the fixed seeds reported in Section VI and run end-to-end on a single CPU core.

XI. APPENDIX: COMPLETE PROOFS OF THEOREMS

A. PROOF OF THEOREM 1: LIPSCHITZ STABILITY OF THE INPUT GRADIENT

Here we provide the detailed algebraic derivation of the input gradient perturbation bound stated in Theorem 1.

Proof. Let the class logit score $S_c(\boldsymbol{x})$ for an input vector $\boldsymbol{x} \in \mathbb{R}^d$ under an L-layer neural network parameterised by weights $\boldsymbol{\theta} = \{\boldsymbol{W}_1, \boldsymbol{W}_2, \dots, \boldsymbol{W}_L\}$ be written as:

$$S_c(\boldsymbol{x}) = \boldsymbol{W}_L^\top \sigma\Big(\boldsymbol{W}_{L-1}\sigma\big(\dots\sigma(\boldsymbol{W}_1\boldsymbol{x})\big)\Big), \tag{23}$$

where σ is a element-wise activation function. The input gradient attribution map $\boldsymbol{g} = \nabla_{\boldsymbol{x}} S_c(\boldsymbol{x})$ is obtained via the chain rule as a product of layer weight matrices and activation derivative Jacobians:

$$\boldsymbol{g} = \boldsymbol{W}_1^\top D_1 \boldsymbol{W}_2^\top D_2 \dots \boldsymbol{W}_{L-1}^\top D_{L-1} \boldsymbol{W}_L, \tag{24}$$

where $D_l = \operatorname{diag}\big(\sigma'(\boldsymbol{z}_l)\big) \in \mathbb{R}^{n_l \times n_l}$ is the diagonal Jacobian of the activation function at layer l, evaluated at the pre-activation vector $\boldsymbol{z}_l$. Since σ is typically a standard activation function (such as ReLU, LeakyReLU, or Sigmoid), its derivative is bounded, satisfying:

$$\|D_l\|_2 = \max_i |\sigma'([\boldsymbol{z}_l]_i)| \leq 1 \quad \forall l \in \{1, \dots, L-1\}. \tag{25}$$

Now consider a perturbation applied to each layer's weights, parameterised as $\tilde{\boldsymbol{W}}_l = \boldsymbol{W}_l + \boldsymbol{E}_l$. The perturbed input gradient is:

$$\tilde{\boldsymbol{g}} = (\boldsymbol{W}_1 + \boldsymbol{E}_1)^\top \tilde{D}_1 (\boldsymbol{W}_2 + \boldsymbol{E}_2)^\top \tilde{D}_2 \dots (\boldsymbol{W}_{L-1} + \boldsymbol{E}_{L-1})^\top \tilde{D}_{L-1} (\boldsymbol{W}_L \tag{26}$$

where $\tilde{D}_l = \operatorname{diag}\big(\sigma'(\tilde{\boldsymbol{z}}_l)\big)$ is the Jacobian evaluated under the perturbed pre-activations.

To bound the difference $\|\tilde{\boldsymbol{g}} - \boldsymbol{g}\|_2$, we decompose the total perturbation into first-order and higher-order terms. First, under the local linear stability assumption (Assumption 1), the active activation regions are assumed to be fixed for small perturbations, meaning $\tilde{D}_l \approx D_l$. Expanding the product in (26) and collecting terms linear in the perturbations $\boldsymbol{E}_l$ gives:

$$\tilde{\boldsymbol{g}} = \boldsymbol{g} + \sum_{l=1}^{L} \boldsymbol{\delta}_l + \mathcal{R}(\{\boldsymbol{E}_k\}), \tag{27}$$

where each $\boldsymbol{\delta}_l$ represents the first-order perturbation term in which only the l-th weight matrix is replaced by its error $\boldsymbol{E}_l$:

$$\boldsymbol{\delta}_l = \boldsymbol{W}_1^\top D_1 \dots \boldsymbol{E}_l^\top D_l \dots \boldsymbol{W}_{L-1}^\top D_{L-1} \boldsymbol{W}_L, \tag{28}$$

and $\mathcal{R}$ is the remainder containing all higher-order cross-terms $\boldsymbol{E}_i \boldsymbol{E}_j$ ($i \neq j$).

Applying the triangle inequality to the first-order approximation:

$$\|\tilde{\boldsymbol{g}} - \boldsymbol{g}\|_2 \leq \sum_{l=1}^{L} \|\boldsymbol{\delta}_l\|_2 + \|\mathcal{R}\|_2 \,. \tag{29}$$

Using the sub-multiplicativity of the spectral norm ($\|\boldsymbol{AB}\|_2 \leq \|\boldsymbol{A}\|_2 \|\boldsymbol{B}\|_2$) and the bound $\|D_k\|_2 \leq 1$, the norm of each individual term $\boldsymbol{\delta}_l$ is bounded by:

$$\begin{aligned}\|\boldsymbol{\delta}_l\|_2 &= \left\|\boldsymbol{W}_1^\top D_1 \dots \boldsymbol{E}_l^\top D_l \dots \boldsymbol{W}_{L-1}^\top D_{L-1} \boldsymbol{W}_L\right\|_2 \\ &\leq \left(\prod_{k\neq l} \|\boldsymbol{W}_k\|_2\right)\left(\prod_{k=1}^{L-1} \|D_k\|_2\right)\|\boldsymbol{E}_l\|_2 \\ &\leq \left(\prod_{k\neq l} \|\boldsymbol{W}_k\|_2\right)\|\boldsymbol{E}_l\|_2 . \end{aligned} \tag{30}$$

Substituting this back into (29) yields the first-order bound:

$$\|\tilde{\boldsymbol{g}} - \boldsymbol{g}\|_2 \leq \sum_{l=1}^{L}\left(\prod_{k\neq l}\|\boldsymbol{W}_k\|_2\right)\|\boldsymbol{E}_l\|_2 + O(\|\Delta\boldsymbol{\theta}\|_2^2). \tag{31}$$

By defining the gradient amplification factor $\Lambda_{\mathrm{S}} = \sum_{l=1}^{L}\prod_{k\neq l}\|\boldsymbol{W}_k\|_2$, and factoring out the maximum relative layer-wise error, we arrive at the desired result:

$$\|\tilde{\boldsymbol{g}} - \boldsymbol{g}\|_2 \leq \Lambda_{\mathrm{S}} \max_l \frac{\|\boldsymbol{E}_l\|_2}{\|\boldsymbol{W}_l\|_2}\|\boldsymbol{W}_l\|_2 + O(\|\Delta\boldsymbol{\theta}\|_2^2). \tag{32}$$

When the activation pattern varies ($\tilde{D}_l \neq D_l$), we bound the difference $\left\|\tilde{D}_l - D_l\right\|_2$. Assuming the activation derivative σ' is β-Lipschitz, we have:

$$\left\|\tilde{D}_l - D_l\right\|_2 \leq \beta\,\|\tilde{\boldsymbol{z}}_l - \boldsymbol{z}_l\|_2 . \tag{33}$$

Since the pre-activation shift $\|\tilde{\boldsymbol{z}}_l - \boldsymbol{z}_l\|_2$ is bounded by $C\,\|\Delta\boldsymbol{\theta}\|_2$ to first order, the perturbation term $\left\|\tilde{D}_l - D_l\right\|_2$ contributes an additional error of order $O(\beta\,\|\Delta\boldsymbol{\theta}\|_2^2)$, which is absorbed directly into the second-order curvature remainder. This completes the proof. □

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